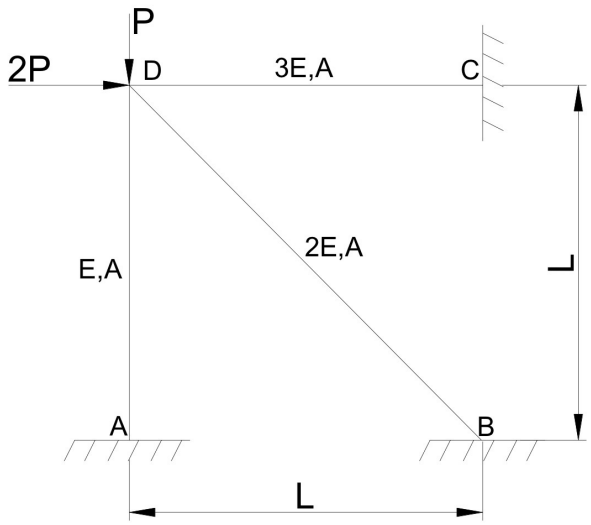
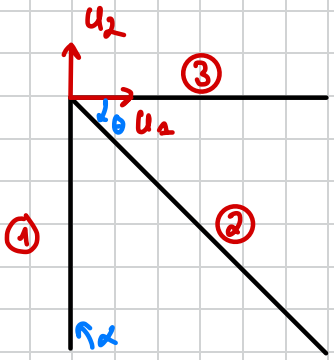
①





2g.l ⇒ K_{2x2}

$L^{①} = L$

$L^{②} = \sqrt{L^2 + L^2} = 1{,}414\,L$

$L^{③} = L$

m.c	1	2	3	4	θ	L
①	-	-	1	2	90°	L
②	1	2	-	-	-45°	1,414L
③	1	2	-	-	0°	L

a) Deslocamentos

$$K_{11} = K_{33}^{①} + K_{11}^{②} + K_{11}^{③} = \frac{EA}{L} \times \cos^2 90° + \frac{2EA}{1{,}414L} \times \cos^2(-45) + \frac{3EA}{L} \cdot \cos^2 0° = \boxed{\frac{EA}{L} \times 3{,}707}$$

$$K_{12} = K_{34}^{①} + K_{12}^{②} + K_{12}^{③} = \frac{EA}{L} \times \sin 90 \cos 90 + \frac{2EA}{1{,}414L} \times \sin(-45)\cos(-45) + \frac{3EA}{L} \sin 0 \cos 0 =$$

$$= \boxed{\frac{EA}{L} \times (-0{,}707)}$$

$$K_{21} = K_{12}$$

$$K_{22} = K_{44}^{①} + K_{22}^{②} + K_{22}^{③} = \frac{EA}{L} \times \sin^2 90 + \frac{2EA}{1{,}414L} \times \sin^2 -45 + \frac{3EA}{L} \times \sin^2 0 = \boxed{\frac{EA}{L} \times 1{,}707}$$

$$[K] = \frac{EA}{L} \begin{bmatrix} 3{,}707 & -0{,}707 \\ -0{,}707 & 1{,}707 \end{bmatrix}$$

$$[K][u] = [F]$$

$$\begin{bmatrix} 3{,}707 & -0{,}707 \\ -0{,}707 & 1{,}707 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 2P \\ -P \end{Bmatrix} \Leftrightarrow \begin{cases} 3{,}707u_1 - 0{,}707u_2 = 2P \\ -0{,}707u_1 + 1{,}707u_2 = -P \end{cases}$$

$$(=) \begin{cases} u_1 = +0{,}4645 \ \frac{PL}{EA} \\ u_2 = -0{,}3934 \ \frac{PL}{EA} \end{cases}$$

b) Esforço Normal

$\sigma = \varepsilon . E$

$\varepsilon = \frac{\delta}{L} \Rightarrow \sigma = \frac{E}{L}(u_2 - u_1)$

$\delta = u_2 - u_1$

$$[u] = [R][u] \Rightarrow \begin{matrix} u_1 \\ u_2 \end{matrix} = \begin{bmatrix} \cos(-45) & \sin(-45) & 0 & 0 \\ 0 & 0 & \cos(-45) & \sin(-45) \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ 0 \\ 0 \end{bmatrix} (=)$$

$$(=) \begin{cases} u_1 = u_1 \times \cos(-45) + u_2 \sin(-45) \\ u_2 = 0 \end{cases} (=) \begin{cases} u_1 = 0{,}607 \ \frac{PL}{EA} \\ u_2 = 0 \end{cases}$$

$$\sigma = \frac{2E}{1{,}414L}\left(0 - 0{,}607 \ \frac{PL}{EA}\right) = -0{,}8586 \ \frac{P}{A}$$

$$\sigma_N = \frac{N}{A} \ (=) \ N = \sigma_N \times A = -0{,}8586 \ \frac{P}{A} \times A = -0{,}8586P$$

c) Modos de vibração

$$[[K] - \omega^2 [M]] = 0$$

$$M_{11} = M_{33}^{(1)} + M_{11}^{(2)} + M_{11}^{(3)} = \frac{\rho AL}{420} \times 140\cos^2 90 + \frac{\rho A 1{,}414L}{420} \times 140\cos^2(-45) + \frac{\rho AL}{420} \times 140\cos^2 0$$

$$= \rho AL\,(0{,}563)$$

$$M_{12} = M_{34}^{(1)} + M_{12}^{(2)} + M_{12}^{(3)} = \frac{\rho AL}{420} \times 140\cos 90 \sin 90 + \frac{\rho A 1{,}414L}{420} \times 140\cos(-45)\sin(-45) +$$

$$+ \frac{\rho AL}{420} \times 140\cos 0 \sin 0 = \rho AL\,(-0{,}236)$$

$$M_{22} = M_{44}^{(1)} + M_{22}^{(2)} + M_{22}^{(3)} = \frac{\rho AL}{420} \times 140\sin^2 90 + \frac{\rho A 1{,}414L}{420} \times 140\sin^2(-45) + \frac{\rho AL}{420} \times 140\sin^2 0 = \rho AL\,(0{,}563)$$

$$\left| \frac{EA}{L} \begin{bmatrix} 3{,}707 & -0{,}1707 \\ -0{,}1707 & 1{,}707 \end{bmatrix} - \omega^2 \rho AL \begin{bmatrix} 0{,}563 & -0{,}236 \\ -0{,}236 & 0{,}563 \end{bmatrix} \right| = 0$$

$$\lambda = \frac{\omega^2 \rho A L \times L}{EA} = \frac{\omega^2 \rho A L^2}{EA} = \frac{\omega^2 \rho L^2}{E}$$

$$\left| \begin{bmatrix} 3{,}707 & -0{,}707 \\ -0{,}707 & 1{,}707 \end{bmatrix} - \lambda \begin{bmatrix} 0{,}563 & -0{,}236 \\ -0{,}236 & 0{,}563 \end{bmatrix} \right| = 0 \Leftrightarrow \left| \begin{bmatrix} 3{,}707 - 0{,}563\lambda & -0{,}707 + 0{,}236\lambda \\ -0{,}707 + 0{,}236\lambda & 1{,}707 - 0{,}563\lambda \end{bmatrix} \right| = 0$$

$$\Leftrightarrow (3{,}707 - 0{,}563\lambda) \times (1{,}707 - 0{,}563\lambda) - (-0{,}707 - 0{,}236\lambda)(-0{,}707 - 0{,}236\lambda) = 0 \Leftrightarrow$$

$$\Leftrightarrow 0{,}268\lambda^2 - 2{,}747\lambda + 5{,}828 = 0 \Leftrightarrow \lambda = \frac{2{,}747 \pm \sqrt{(-2{,}747)^2 - 4 \times 0{,}268 \times 5{,}828}}{2 \times 0{,}268} \Leftrightarrow$$

$$\Leftrightarrow \lambda_1 = 3 \quad \wedge \quad \lambda_2 = 7{,}25$$

$$\omega_1 = \sqrt{\frac{\lambda E}{\rho L^2}} = \sqrt{\frac{3E}{\rho L^2}} = \frac{1{,}73}{L}\sqrt{\frac{E}{\rho}}$$

$$\omega_2 = \sqrt{\frac{7{,}24E}{\rho L^2}} = \frac{2{,}69}{L}\sqrt{\frac{E}{\rho}}$$

Associado a λ_1: ω_1

$$\begin{bmatrix} 3{,}707 - 3(0{,}563) & -0{,}707 + 0{,}236(3) \\ -0{,}707 + 0{,}236(3) & 1{,}707 - 0{,}563(3) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = 0 \Leftrightarrow \begin{cases} 2u_1 + 0{,}001u_2 = 0 \\ 0{,}001u_1 + 0u_2 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = \text{qualquer} \end{cases}$$

$u_2 = 1 \quad \wedge \quad u_1 = 0$



Associado a λ_2: ω_2

$$\begin{bmatrix} 3{,}707 - 7{,}24(0{,}563) & -0{,}707 + 0{,}236(7{,}24) \\ -0{,}707 + 0{,}236(7{,}24) & 1{,}707 - 0{,}563(7{,}24) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = 0 \Leftrightarrow \begin{cases} -0{,}413u_1 + 1u_2 = 0 \\ 1u_1 - 2{,}41u_2 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 2{,}43\,u_2 \\ u_1 = 2{,}41\,u_2 \end{cases} \Rightarrow \begin{cases} u_1 \wedge u_2 = 0 \\ u_1 \approx 2{,}42\,u_2 \end{cases}$$

$u_1 = 2{,}43$
$u_2 = 1$



②

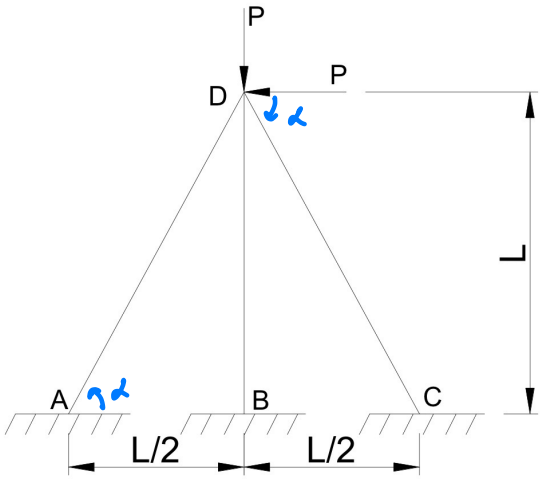
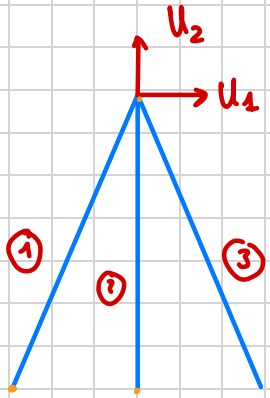


Figura 2



a) Deslocamentos ponto D

$L^{①} = \sqrt{(L/2)^2 + L^2} = 1{,}118L$

$L^{②} = L$

$L^{③} = 1{,}118L$

$\alpha = tg^{-1}\left(\frac{L}{L/2}\right) = tg^{-1}(2) = 63{,}435°$

m.c

	1	2	3	4	θ	L
①	-	-	1	2	63,435	1,118L
②	-	-	1	2	90°	L
③	1	2	-	-	-63,435	1,118L

2g.l ⇒ K_{2x2}

$K_{11} = K_{33}^{①} + K_{33}^{②} + K_{11}^{③} = \frac{EA}{1{,}118L} \times \cos^2 63{,}435 + \frac{EA}{L}\cos^2 90^0 + \frac{EA}{1{,}118L} \times \cos^2(-63{,}435)$

$= \frac{EA}{L} \times (0{,}35778)$ //

$K_{12} = K_{34}^{①} + K_{34}^{②} + K_{12}^{③} = \frac{EA}{1{,}118L} \times \sin 63{,}435 \cos 63{,}435 + \frac{EA}{L} \times \sin 90 \cos 90^0 + \frac{EA}{1{,}118L} \cos -63{,}435 \sin -63{,}435$

$= \frac{EA}{L} \times 0 = 0$ //

$K_{22} = K_{44}^{①} + K_{44}^{②} + K_{22}^{③} = \frac{EA}{1{,}118L} \times \sin^2 63{,}435 + \frac{EA}{L} \times \sin^2 90 + \frac{EA}{1{,}118L} \times \sin^2(-63{,}435) = \frac{EA}{L} \times 2{,}4311$

$$[K] = \frac{EA}{L}\begin{bmatrix} 0{,}35778 & 0 \\ 0 & 2{,}4311 \end{bmatrix}$$

$[K]\,[u] = [F]$

$$\frac{EA}{L}\begin{bmatrix} 0{,}35778 & 0 \\ 0 & 2{,}4311 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} -P \\ -P \end{Bmatrix} \Leftrightarrow \begin{matrix} 0{,}35778\frac{EA}{L}u_1 + 0u_2 = -P \\ 0u_1 + 2{,}4311\frac{EA}{L}u_2 = -P \end{matrix} \Leftrightarrow \begin{matrix} u_1 = -2{,}79\frac{PL}{EA} \text{ //} \\ u_2 = -0{,}411\frac{PL}{EA} \text{ //} \end{matrix}$$

b) Tensão axial DC

$\sigma = E.\varepsilon$

$\varepsilon = \frac{\delta}{L} \Rightarrow \sigma = \frac{E}{L} \times (u_2 - u_1)$

$\delta = u_2 - u_1$

$\theta = -63{,}435°$

$[u] = [R][q]$

$$[u] = \begin{bmatrix} \cos -63{,}435 & \sin \cdot 63{,}435 & 0 & 0 \\ 0 & 0 & \cos \cdot 63{,}435 & \sin \cdot 63{,}435 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ 0 \\ 0 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = u_1 \times \cos -63{,}435° + u_2 \times \sin \cdot 63{,}435° \\ u_2 = 0 \end{cases} \Leftrightarrow \begin{cases} u_1 = -0{,}88 \frac{PL}{EA} \\ u_2 = 0 \end{cases}$$

$$\sigma = \frac{E}{1{,}118L} \times (0 - (-0{,}88 \frac{PL}{EA})) \Leftrightarrow \sigma = +0{,}787 \frac{P}{A}$$

c) Modos de vibração

$|[K] - \omega^2 [m]| = 0$

$M_{11} = M_{33}^{①} + M_{33}^{②} + M_{11}^{③} = \frac{\rho A.1.118L}{420} \times 140 \cos^2 63{,}435 + \frac{\rho AL}{420} \times 140 \cos^2 90° +$

$+ \frac{\rho A.1.118L}{420} \times 140 \cos^2 (-63{,}435) = \rho AL \times 0{,}075$ //

$M_{12} = M_{34}^{①} + M_{34}^{②} + M_{12}^{③} = \frac{\rho A.1.118L}{420} \times 140 \times \sin 63{,}435 \cos 63{,}435 + \frac{\rho AL}{420} \times 140 \times \sin 90 \cos 90 +$

$\frac{\rho A.1.118L}{420} \times 140 \cos -63{,}435 \sin \cdot 63{,}435 = \rho AL \times 0$ //

$M_{22} = M_{44}^{①} + M_{44}^{②} + M_{22}^{③} = \frac{\rho A.1.118L}{420} \times 140 \sin^2 63{,}435 + \frac{\rho AL}{420} \times 140 \sin^2 90° + \frac{\rho A.1.118L}{420} \times 140 \sin^2 -63{,}435$

$= 0{,}634 \ \rho AL$ //

Erro de cálculo talvez porém raciocínio certo

$$\left| \frac{EA}{L} \begin{bmatrix} 0{,}35518 & 0 \\ 0 & 2{,}4311 \end{bmatrix} - \omega^2 \rho AL \begin{bmatrix} 0{,}075 & 0 \\ 0 & 0{,}634 \end{bmatrix} \right| = 0$$

$$\lambda = \frac{\omega^2 \rho AL \times L}{EA} = \frac{\omega^2 \rho L^2}{E}$$

$$\left| \begin{bmatrix} 0,35578 & 0 \\ 0 & 2,4311 \end{bmatrix} - \lambda \begin{bmatrix} 0,035 & 0 \\ 0 & 0,634 \end{bmatrix} \right| \overset{\neq 0}{=} 0 \Leftrightarrow \left| \begin{bmatrix} 0,35578 - 0,035\lambda & 0 \\ 0 & 2,4311 - 0,634\lambda \end{bmatrix} \right| = 0$$

$$\Leftrightarrow (0,35578 - 0,035\lambda)(2,4311 - 0,634\lambda) - (0) \times (0) = 0 \Leftrightarrow$$

$$\Leftrightarrow 0,04435\lambda^2 - 0,4079\lambda + 0,865 = 0 \Leftrightarrow \lambda = \frac{0,4079 \pm \sqrt{(0,4079^2) - 4 \times 0,04435 \times 0,865}}{2 \times 0,04435}$$

$$\Leftrightarrow \lambda_1 = 3,84 \quad \wedge \quad \lambda_2 = 4,74$$

$$w_1 = \sqrt{\frac{\lambda E}{\rho L^2}} = \sqrt{\frac{3,84 E}{\rho L^2}} = \frac{1,96}{L} \times \sqrt{\frac{E}{\rho}}$$

$$w_2 = \sqrt{\frac{4,74 E}{\rho L^2}} = \frac{2,178}{L} \times \sqrt{\frac{E}{\rho}}$$

Associado a $\lambda_1 : w_1$

$$\begin{bmatrix} 0,35578 - 0,035(3,84) & 0 \\ 0 & 2,4311 - 0,634(3,84) \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \end{Bmatrix} = 0 \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 0,06778 U_1 + 0 U_2 = 0 \\ 0 U_1 - 0,00346 U_2 = 0 \end{cases} \Leftrightarrow \begin{cases} U_1 = 0 \\ U_2 = 0 \end{cases} \text{ qualquer}$$

$U_1 = 0 \quad \wedge \quad U_2 = 1$



Associado a $\lambda_2 : w_2$

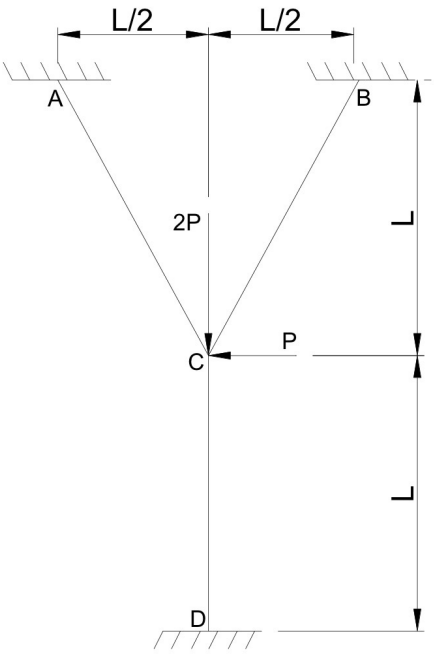
$$\begin{bmatrix} 0,35578 - 0,035(4,74) & 0 \\ 0 & 2,4311 - 0,634(4,74) \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \end{Bmatrix} = 0 \Leftrightarrow$$

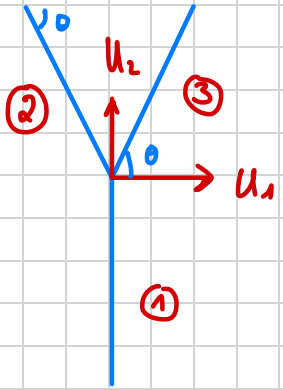
$$\Leftrightarrow \begin{cases} 0,00028 U_1 + 0 U_2 = 0 \\ 0 U_1 - 0,57406 U_2 = 0 \end{cases} \Leftrightarrow \begin{cases} U_1 = 0 \\ U_2 = 0 \end{cases} \text{ qualquer}$$

$U_1 = 1 \quad U_2 = 0 \Rightarrow$



③





$2gl \Rightarrow [K]_{2\times 2}$

$L^{①} = L$

$L^{②} = \sqrt{(L/2)^2 + (L)^2} = 1{,}118L$

$L^{③} = 1{,}118L$

$\theta = tg^{-1}\left(\frac{L}{L/2}\right) = tg^{-1}(2) = 63{,}435^\circ$

m.c

	1	2	3	4	θ	L
①	-	-	1	2	90°	L
②	-	-	1	2	-63,435	1,118L
③	1	2	-	-	63,435°	1,118L

a) Deslocamentos em C

$K_{11} = K_{33}^{①} + K_{33}^{②} + K_{11}^{③} = \frac{EA}{L} \times \cos^2 90^\circ + \frac{EA}{1{,}118L} \times \cos^2 -63{,}435^\circ + \frac{EA}{1{,}118L} \times \cos^2 63{,}435 =$

$= 0{,}35778 \frac{EA}{L}$ //

$K_{12} = K_{34}^{①} + K_{34}^{②} + K_{12}^{③} = \frac{EA}{L} \times \sin 90 \cos 90^\circ + \frac{EA}{1{,}118L} \times \sin -63{,}435 \cos 63{,}435 + \frac{EA}{1{,}118L} \times \sin 63{,}435 \cos 63{,}435^\circ$

$= 0 \frac{EA}{L}$ //

$K_{22} = K_{44}^{①} + K_{44}^{②} + K_{22}^{③} = \frac{EA}{L} \times \sin^2 90 + \frac{EA}{1{,}118L} \times \sin^2(-63{,}435) + \frac{EA}{1{,}118L} \times \sin^2(63{,}435) =$

$= \frac{EA}{L} \times 2{,}4311$ //

$$[K] = \frac{EA}{L}\begin{bmatrix} 0{,}35778 & 0 \\ 0 & 2{,}4311 \end{bmatrix}$$

$$[K][u] = [F] \Rightarrow \frac{EA}{L}\begin{bmatrix} 0{,}35778 & 0 \\ 0 & 2{,}4311 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} -P \\ -2P \end{Bmatrix} \Leftrightarrow \begin{matrix} 0{,}35778 \frac{EA}{L} u_1 = -P \\ 2{,}4311 \frac{EA}{L} u_2 = -2P \end{matrix} \Rightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = -2{,}79 \frac{PL}{EA} \\ u_2 = -0{,}82 \frac{PL}{EA} \end{cases}$$

b) Tensão axial em AC

$\theta = -63{,}435$

$\sigma = E.\varepsilon$

$\varepsilon = \frac{\delta}{L} \Rightarrow \sigma = \frac{E}{L}(u_2 - u_1)$

$\delta = u_2 - u_1$

$$[u] = [R][u] \Leftrightarrow \begin{matrix} u_1 \\ u_2 \end{matrix} \left\{ = \begin{bmatrix} \cos -63{,}435 & \sin -63{,}435 & 0 & 0 \\ 0 & 0 & \cos -63{,}435 & \sin -63{,}435 \end{bmatrix} \right\} \begin{matrix} 0 \\ 0 \\ u_1 \\ u_2 \end{matrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = u_1 \cos -63{,}435 + u_2 \sin -63{,}435 \end{cases} \Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = -0{,}515 \frac{PL}{EA} \end{cases}$$

$$\sigma = \frac{E}{1{,}118L} \times (-0{,}515 - 0)\frac{PL}{EA} \Rightarrow \sigma = -0{,}4606 \frac{P}{A}$$

c) Modos de vibração

$$|[K] - \omega^2 [\pi]| = 0$$

$$M_{11} = M_{33}^{①} + M_{33}^{②} + M_{11}^{③} = \frac{\rho A.L}{420} \times 140 \cos^2 90 + \frac{\rho A.1{,}118L}{420} \times 140 \cos^2(-63{,}435) + \frac{\rho A.1{,}118L}{420} \times 140 \times \cos^2 63{,}435$$

$$= \rho A.L \times 0{,}149$$

$$M_{12} = M_{34}^{①} + M_{34}^{②} + M_{12}^{③} = \frac{\rho A.L}{420} \times 140 \times \sin 90 \cos 90 + \frac{\rho A.1{,}118L}{420} \times 140 \times \sin(-63{,}435)\cos(63{,}435) +$$

$$+ \frac{\rho A.1{,}118L}{420} \times 140 \times \sin 63{,}435 \cos 63{,}435 = \rho AL \times 0$$

$$M_{22} = M_{44}^{①} + M_{44}^{②} + M_{22}^{③} = \frac{\rho A.L}{420} \times 140 \sin^2 90 + \frac{\rho A.1{,}118L}{420} \times 140 \sin^2 -63{,}435 + \frac{\rho A.1{,}118L}{420} \times 140 \sin^2 63{,}435 =$$

$$= \rho AL \times 0{,}93$$

$$\left| \frac{EA}{L} \begin{bmatrix} 0{,}35538 & 0 \\ 0 & 2{,}4311 \end{bmatrix} - \omega^2 \rho AL \begin{bmatrix} 0{,}149 & 0 \\ 0 & 0{,}93 \end{bmatrix} \right| = 0, \quad \lambda = \frac{\omega^2 \rho A.L.L}{EA} = \frac{\omega^2 \rho L^2}{E}$$

$$\left| \begin{bmatrix} 0{,}35538 & 0 \\ 0 & 2{,}4311 \end{bmatrix} - \lambda \begin{bmatrix} 0{,}149 & 0 \\ 0 & 0{,}93 \end{bmatrix} \right| = 0 \iff$$

$$\iff \left| \begin{bmatrix} 0{,}35538 - 0{,}149\lambda & 0 \\ 0 & 2{,}4311 - 0{,}93\lambda \end{bmatrix} \right| = 0 \iff (0{,}35538 - 0{,}149\lambda)(2{,}4311 - 0{,}93\lambda) = 0$$

$$\iff 0{,}138571\lambda^2 - 0{,}693\lambda + 0{,}865 = 0 \iff \lambda = \frac{0{,}693 \pm \sqrt{(-0{,}693)^2 - 4 \times 0{,}138571 \times 0{,}865}}{2 \times 0{,}138571} \iff$$

$$\iff \lambda_1 = 2{,}40 \quad \wedge \quad \lambda_2 = 2{,}60$$

$$\omega_1 = \sqrt{\frac{\lambda E}{\rho L^2}} = \sqrt{\frac{2{,}40\,E}{\rho L^2}} = \frac{1{,}54}{L} \times \sqrt{\frac{E}{\rho}}$$

$$\omega_2 = \sqrt{\frac{2{,}60E}{\rho L^2}} = \frac{1{,}61}{L} \times \sqrt{\frac{E}{\rho}} \quad //$$

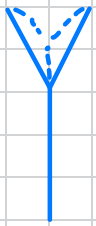
Associado a λ_1 : ω_1

$$\begin{bmatrix} 0{,}35538 - 0{,}149(2{,}40) & 0 \\ 0 & 2{,}4311 - 0{,}93(2{,}40) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = 0 \iff \begin{cases} -0{,}00182\,u_1 = 0 \\ 0{,}1991\,u_2 = 0 \end{cases} \iff$$

$$\iff \begin{cases} u_1 = 0 \\ u_2 = 0 \end{cases}, \text{ qualquer}$$

$u_1 = 0$

$u_2 = 1$



$$\begin{bmatrix} 0{,}35538 - 0{,}149(2{,}60) & 0 \\ 0 & 2{,}4311 - 0{,}93(2{,}60) \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = 0 \iff \begin{cases} -0{,}0316 \cdot u_1 = 0 \\ 0{,}0131\ u_2 = 0 \end{cases} \iff$$

$$\iff \begin{cases} u_1 = 0 \\ u_2 = 0 \end{cases}, \text{ qualquer}$$

$u_1 = 1$
$u_2 = 0$ $\Rightarrow$



④

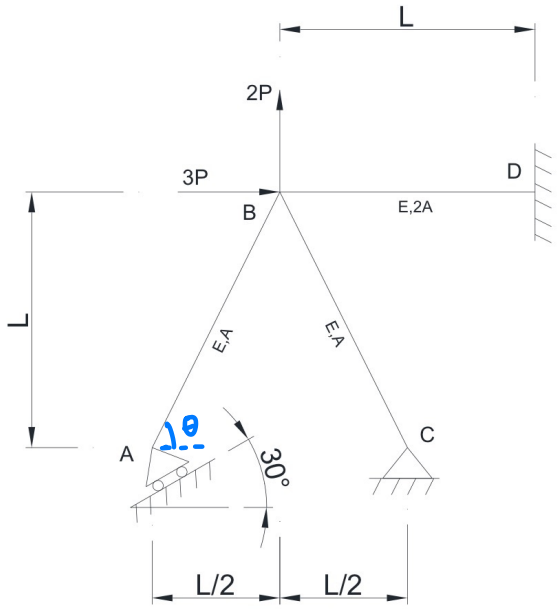
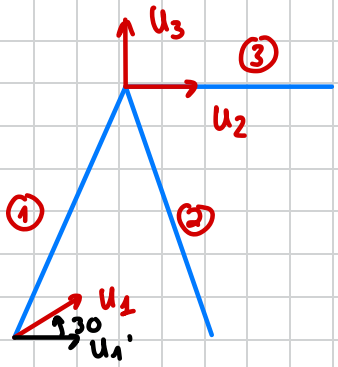


Figura 4



3g.l ⇒ $K_{3\times3}$

$L^{①} = \sqrt{L^2 + (L/2)^2} = 1{,}118L$

$L^{②} = 1{,}118L$

$L^{③} = L$

$\theta = tg^{-1}\left(\frac{L}{L/2}\right) = tg^{-1}(2) = 63{,}435°$

a) Deslocamentos nodais

m.c

	1	2	3	4	θ_1	θ_2	L
①	1	-	2	3	33,435	63,435	1,118L
②	2	3	-	-	-63,435	-63,435	1,118L
③	2	3	-	-	0	0	L

$K_{11} = K_{11}^{①} = \frac{EA}{1{,}118L} \times \cos^2 33{,}435 = \frac{EA}{L} \times 0{,}624$ //

$K_{12} = K_{13}^{①} = \frac{EA}{1{,}118L} \times -\cos 33{,}435 \cos 63{,}435 = \frac{EA}{L} \times (-0{,}33)$ //

$K_{13} = K_{14}^{①} = \frac{EA}{1{,}118L} \times -\sin 63{,}435 \cos 33{,}435 = \frac{EA}{L} \times (-0{,}668)$ //

$K_{22} = K_{33}^{①} + K_{11}^{②} + K_{11}^{③} = \frac{EA}{1{,}118L} \times \cos^2 63{,}435 + \frac{EA}{1{,}118L} \times \cos^2 -63{,}435 + \frac{E2A}{L} \times \cos^2 0° = \frac{EA}{L} \times 2{,}36$ //

$K_{23} = K_{34}^{①} + K_{12}^{②} + K_{12}^{③} = \frac{EA}{1{,}118L} \times \sin 63{,}435 \cos 63{,}435 + \frac{EA}{1{,}118L} \times \sin -63{,}435 \cos 63{,}435 +$

$+ \frac{2EA}{L} \times \sin 0 \cos 0 = \frac{EA}{L} \times 0$ //

$K_{33} = K_{44}^{①} + K_{22}^{②} + K_{22}^{③} = \frac{EA}{1{,}118L} \times \sin^2 63{,}435 + \frac{EA}{1{,}118L} \times \sin^2 -63{,}435 + \frac{2EA}{L} \times \sin^2 0 =$

$= \frac{EA}{L} \times 1{,}431$ //

$$[K] = \frac{EA}{L}\begin{bmatrix} 0,624 & -0,33 & -0,668 \\ -0,33 & 2,36 & 0 \\ -0,668 & 0 & 1,431 \end{bmatrix}$$

$$[K][u] = [F] \Leftrightarrow \frac{EA}{L}\begin{bmatrix} 0,624 & -0,33 & -0,668 \\ -0,33 & 2,36 & 0 \\ -0,668 & 0 & 1,431 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 3P \\ 2P \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 0,624\frac{EA}{L}u_1 - 0,33\frac{EA}{L}u_1 - 0,668\frac{EA}{L}u_3 = 0 \\ -0,33\frac{EA}{L}u_1 + 2,36\frac{EA}{L}u_2 = 3P \\ -0,668\frac{EA}{L}u_1 + 1,431\frac{EA}{L}u_3 = 2P \end{cases} \Leftrightarrow \begin{cases} u_1 = 5,08\ \frac{PL}{EA} \\ u_2 = 1,98\ \frac{PL}{EA} \\ u_3 = 3,77\ \frac{PL}{EA} \end{cases}$$

b) Zona crítica da estrutura

$$\sigma = E.\varepsilon$$

$$\varepsilon = \frac{\delta}{L} \Rightarrow \sigma = \frac{E}{L}(u_2 - u_1)$$

$$\delta = u_2 - u_1$$

Barra AB

$$[u] = [R][u] \Leftrightarrow \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos 33,435 & \sin 33,435 & 0 & 0 \\ 0 & 0 & \cos 63,435 & \sin 63,435 \end{bmatrix}\begin{Bmatrix} u_1 \\ 0 \\ u_2 \\ u_3 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = u_1 \times \cos 33,435 \\ u_2 = u_2 \times \cos 63,435 + u_3 \times \sin 63,435 \end{cases} \Leftrightarrow \begin{cases} u_1 = 4,24\ \frac{PL}{EA} \\ u_2 = 4,25\ \frac{PL}{EA} \end{cases}$$

$$\sigma = \frac{E}{1,118L} \times (4,25 - 4,24)\ \frac{PL}{EA} \Leftrightarrow \sigma = 0,008945\ \frac{P}{A}$$

Barra CB

$$\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos -63,435 & \sin -63,435 & 0 & 0 \\ 0 & 0 & \cos -63,435 & \sin -63,435 \end{bmatrix}\begin{Bmatrix} u_2 \\ u_3 \\ 0 \\ 0 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = u_2 \times \cos -63,435 + u_3 \times \sin -63,435 \\ u_2 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = -2,486\ \frac{PL}{EA} \\ u_2 = 0 \end{cases}$$

$$\sigma = \frac{E}{1,118L} \times (0 - (-2,486))\ \frac{PL}{EA} \Leftrightarrow \sigma = 2,22\ \frac{P}{A}$$

Barra BD

$$\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos 0 & \sin 0 & 0 & 0 \\ 0 & 0 & \cos 0 & \sin 0 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \\ 0 \\ 0 \end{Bmatrix} \Leftrightarrow \begin{cases} u_1 = u_2 \times \cos 0 + u_3 \cancel{\sin 0}^{\,0} \\ u_2 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 1{,}98 \frac{PL}{EA} \\ u_2 = 0 \end{cases}$$

$$\sigma = \frac{E}{L}(0 - 1{,}98)\frac{PL}{EA} \Leftrightarrow \sigma = -1{,}98 \frac{P}{A}$$

c) Deslocamento aproximado barra AB a L/2 de A.

$$u(x) = N_1 . u_1 + N_2 . u_2$$

$$u(x) = \frac{1{,}118L - x}{1{,}118L} . 4{,}24 \frac{PL}{EA} + \frac{x}{1{,}118L} \times 4{,}25 \frac{PL}{EA} \Leftrightarrow$$

$$\Leftrightarrow u(x) = 1 - \frac{x}{1{,}118L} \times 4{,}24 \frac{PL}{EA} + \frac{x}{1{,}118L} \times 4{,}25 \frac{PL}{EA} //$$

$$u(x = L/2) = 1 - \frac{L/2}{1{,}118L} \times 4{,}24 \frac{PL}{EA} + \frac{L/2}{1{,}118L} \times 4{,}25 \frac{PL}{EA} = 4{,}24 \frac{PL}{EA} //$$

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⑤

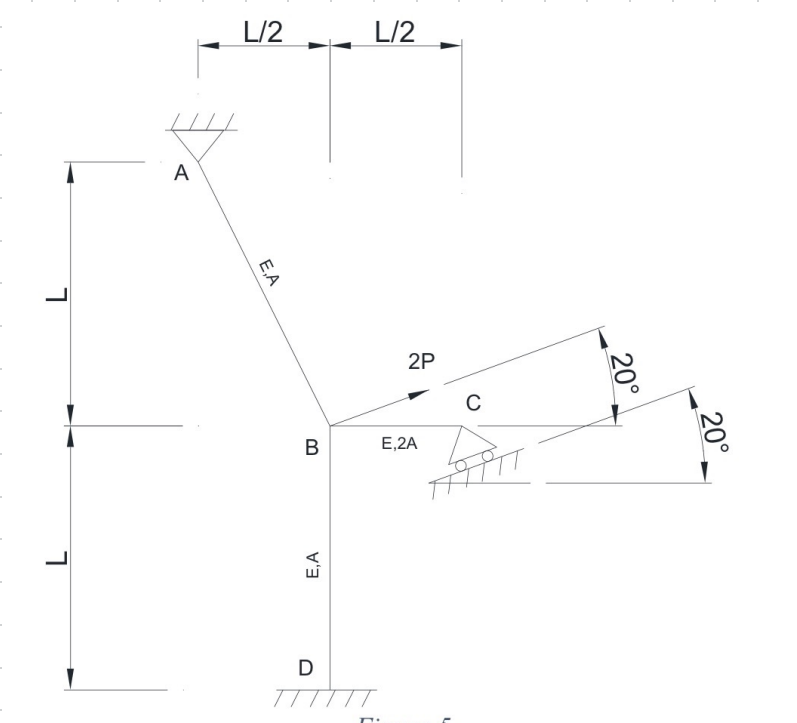
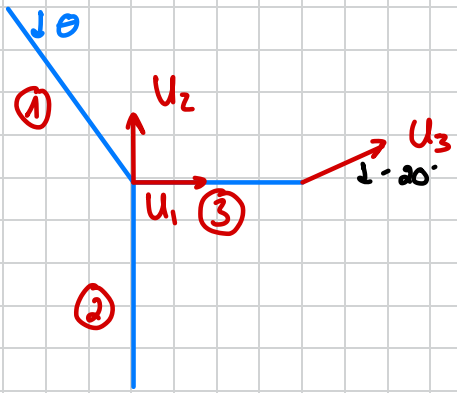


Figura 5



3g.l ⇒ $K_{3\times 3}$

$L^{①} = \sqrt{(L/2)^2 + L^2} = 1{,}118L$

$L^{②} = L$

$L^{③} = L/2$

$\theta = tg^{-1}\left(\frac{L}{L/2}\right) = tg^{-1}(2) = 63,435^\circ$

a) Desloc. nodais

m.c

	1	2	3	4	θ_1	θ_2	L
①	-	-	1	2	-63,435	-63,435	1,118L
②	-	-	1	2	90°	90°	L
③	1	2	3	-	0°	-20°	L/2

$K_{11} = K_{33}^{①} + K_{33}^{②} + K_{11}^{③} = \frac{EA}{1,118L} \times \cos^2 -63,435^\circ + \frac{EA}{L} \times \cos^2 90^\circ + \frac{2EA}{L/2} \times \cos^2 0^\circ = \frac{EA}{L} \times 4,179$ //

$K_{12} = K_{34}^{①} + K_{34}^{②} + K_{12}^{③} = \frac{EA}{1,118L} \times \sin -63,435 \cos -63,435 + \frac{EA}{L} \times \sin 90 \cos 90^\circ + \frac{2EA}{L/2} \times \sin 0 \cos 0 =$

$= \frac{EA}{L} \times (-0,358)$ //

$K_{13} = K_{13}^{③} = \frac{2EA}{L/2} \times (-\cos 0^\circ \cos -20^\circ) = \frac{EA}{L} \times (-3,759)$ //

$K_{22} = K_{44}^{①} + K_{44}^{②} + K_{22}^{③} = \frac{EA}{1,118L} \times \sin^2 -63,435 + \frac{EA}{L} \times \sin^2 90 + \frac{2EA}{L/2} \times \sin^2 0^\circ = \frac{EA}{L} \times 1,716$ //

$K_{23} = K_{23}^{③} = \frac{2EA}{L/2} \times (-\sin 0^\circ \cos -20) = 0$

$K_{33} = K_{33}^{③} = \frac{2EA}{L/2} \times \cos^2 -20 = \frac{EA}{L} \times 3,532$ //

$$[K] = \frac{EA}{L} \begin{bmatrix} 4,179 & -0,358 & -3,759 \\ -0,358 & 1,716 & 0 \\ -3,759 & 0 & 3,532 \end{bmatrix}$$

$[K]\ [u] = [F]$

$$\frac{EA}{L} \begin{bmatrix} 4,179 & -0,358 & -3,759 \\ -0,358 & 1,716 & 0 \\ -3,759 & 0 & 3,532 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} 2P\cos 20^\circ \\ 2P\sin 20^\circ \\ 0 \end{Bmatrix} \Rightarrow$$

$$\Leftrightarrow \begin{cases} 4,179 \frac{EA}{L} \cdot u_1 - 0,358 \frac{EA}{L} u_2 - 3,759 \frac{EA}{L} \cdot u_3 = 2P\cos 20^\circ \\ -0,358 \frac{EA}{L} \cdot u_1 + 1,716 \frac{EA}{L} \cdot u_2 = 2P \cdot \sin 20^\circ \\ -3,759 \frac{EA}{L} \cdot u_1 + 3,532 \frac{EA}{L} u_3 = 0 \end{cases} \Leftrightarrow \begin{cases} u_1 = 19,495 \frac{PL}{EA} \\ u_2 = 4,466 \frac{PL}{EA} \\ u_3 = 20,748 \frac{PL}{EA} \end{cases}$$ //

b) $u(x) = N_1 \cdot u_1 + N_2 \cdot u_2$ (Desl. aprox. a L/2 de A)

$[u] = [R]\ [u]$

$$\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos -63,435 & \sin -63,435 & 0 & 0 \\ 0 & 0 & \cos -63,435 & \sin -63,435 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ u_1 \\ u_2 \end{Bmatrix} \Rightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = u_1 \times \cos -63{,}435 + u_2 \times \sin \cdot 63{,}435 \end{cases} \Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = 4{,}724 \ \frac{PL}{EA} \end{cases}$$

$$u(x) = \frac{1{,}118L - x}{1{,}118L} \times 0 + \frac{x}{1{,}118L} \times 4{,}724 \ \frac{PL}{EA} \Leftrightarrow u(x) = \frac{x}{1{,}118L} \times 4{,}724 \ \frac{PL}{EA}$$

$$u(x = L/2) = \frac{(L/2)}{1{,}118L} \times 4{,}724 \ \frac{PL}{EA} = 2{,}11 \ \frac{PL}{EA} \ //$$

c) Zona crítica

$$\sigma = E \cdot \varepsilon$$

$$\varepsilon = \frac{\delta}{L} \Rightarrow \sigma = \frac{E}{L}(u_2 - u_1)$$

$$\delta = u_2 - u_1$$

Barra AB

$$\sigma = \frac{E}{1{,}118L} \times (4{,}724 - 0) \frac{PL}{EA} = 4{,}225 \ \frac{P}{A}$$

Barra BD

$$[u] = [R][u] \Leftrightarrow \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos 90 & \sin 90 & 0 & 0 \\ 0 & 0 & \cos 90 & \sin 90 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ u_1 \\ u_2 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = u_1 \times \cos 90^\circ + u_2 \times \sin 90^\circ \end{cases} \Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = 4{,}466 \ \frac{PL}{EA} \end{cases}$$

$$\sigma = \frac{E}{L} \times (4{,}466 - 0) \frac{PL}{EA} = 4{,}466 \ \frac{P}{A}$$

Barra BC

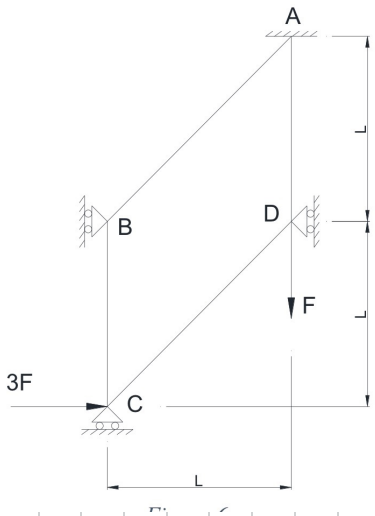
$$[u] = [R][u] \Leftrightarrow \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos 0 & \sin 0 & 0 & 0 \\ 0 & 0 & \cos \cdot 20 & \sin \cdot 20 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ 0 \end{Bmatrix} \Leftrightarrow$$

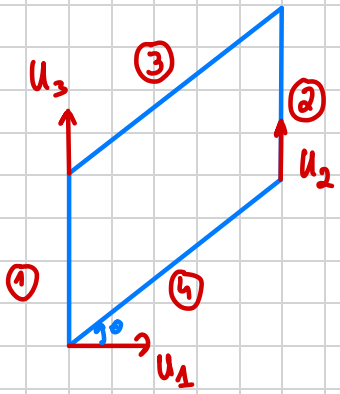
$$\Leftrightarrow \begin{cases} u_1 = u_1 \times \cos 0 + u_2 \times \sin 0^\circ \\ u_2 = u_3 \cos \cdot 20^\circ \end{cases} \Leftrightarrow \begin{cases} u_1 = 19{,}495 \ \frac{PL}{EA} \\ u_2 = 19{,}497 \ \frac{PL}{EA} \end{cases}$$

$$\sigma = \frac{E}{L/2} \times (19{,}497 - 19{,}495) \frac{PL}{EA} = 0{,}004 \ \frac{P}{A}$$

R: Zona crítica na barra BD.

⑥





3g.l ⇒ $[K]_{3\times3}$

a) Deslocamentos nodais

$L^{①} = L$

$L^{②} = L$

$L^{③} = \sqrt{L^2+L^2} = 1{,}414L$

$L^{④} = 1{,}414L$

$\theta = tg^{-1}\left(\frac{L}{L}\right) = tg^{-1}(1) = 45°$

m.c

	1	2	3	4	θ	L
①	1	-	-	3	90°	L
②	-	2	-	-	90°	L
③	-	3	-	-	45°	1,414L
④	1	-	-	2	45	1,414L

$K_{11} = K_{11}^{①} + K_{11}^{④} = \frac{EA}{L} \times \cos^2 90 + \frac{EA}{1{,}414L} \times \cos^2 45 = \frac{EA}{L} \times 0{,}354$ //

$K_{12} = K_{14}^{④} = \frac{EA}{1{,}414L} \times (-\sin 45 \cos 45) = \frac{EA}{L} \times (-0{,}354)$ //

$K_{13} = K_{14}^{①} = \frac{EA}{L} \times (-\sin 90 \cos \cancel{90}^{0}) = 0$ //

$K_{22} = K_{22}^{②} + K_{44}^{④} = \frac{EA}{L} \times \sin^2 90 + \frac{EA}{1{,}414L} \times \sin^2 45 = \frac{EA}{L} \times 1{,}354$ //

$K_{23} = 0$

$K_{33} = K_{44}^{①} + K_{22}^{③} = \frac{EA}{L} \times \sin^2 90 + \frac{EA}{1{,}414L} \times \sin^2 45 = \frac{EA}{L} \times 1{,}354$ //

$$[K] = \frac{EA}{L}\begin{bmatrix} 0{,}354 & -0{,}354 & 0 \\ -0{,}354 & 1{,}354 & 0 \\ 0 & 0 & 1{,}354 \end{bmatrix}$$

$$[K][u] = [F]$$

$$\frac{EA}{L}\begin{bmatrix} 0{,}354 & -0{,}354 & 0 \\ -0{,}354 & 1{,}354 & 0 \\ 0 & 0 & 1{,}354 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} 3F \\ -F \\ 0 \end{Bmatrix} \Leftrightarrow \begin{cases} 0{,}354\frac{EA}{L}u_2 - 0{,}354\frac{EA}{L}\cdot u_2 = 3F \\ -0{,}354\frac{EA}{L}\cdot u_1 + 1{,}354\frac{EA}{L}\cdot u_2 = -F \\ 1{,}354 u_3 = 0 \end{cases} \Leftrightarrow \begin{cases} u_1 = 10{,}434\frac{FL}{EA} \\ u_2 = 2\frac{FL}{EA} \\ u_3 = 0 \end{cases}$$

b) Tensão axial CD

$$[u] = [R][u] \Leftrightarrow \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos 45 & \sin 45 & 0 & 0 \\ 0 & 0 & \cos 45 & \sin 45 \end{bmatrix}\begin{Bmatrix} u_1 \\ 0 \\ 0 \\ u_2 \end{Bmatrix} \Leftrightarrow \begin{cases} u_1 = u_1 \times \cos 45 \\ u_2 = u_2 \times \sin 45 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = 7{,}406\frac{FL}{EA} \\ u_2 = 1{,}414\frac{FL}{EA} \end{cases}$$

$$\sigma = E.\varepsilon$$

$$\varepsilon = \frac{\delta}{L} \Rightarrow \sigma = \frac{E}{L} \times (u_2 - u_1)$$

$$\delta = u_2 - u_1$$

$$\sigma = \frac{E}{1{,}414\frac{L}{}} \times (1{,}414 - 7{,}406)\frac{FL}{EA} = -4{,}24\frac{P}{A}$$

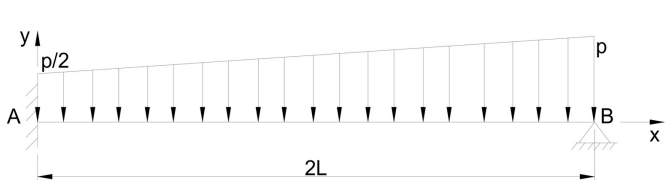
c) Desl. aproximado a meio banna AB

$$[u] = [R][u] \Leftrightarrow \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{bmatrix} \cos 45 & \sin 45 & 0 & 0 \\ 0 & 0 & \cos 45 & \sin 45 \end{bmatrix}\begin{Bmatrix} 0 \\ u_3 \\ 0 \\ 0 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = \overset{0}{u_3} \times \sin 45 \\ u_2 = 0 \end{cases} \Leftrightarrow \begin{cases} u_1 = 0 \\ u_2 = 0 \end{cases} \Rightarrow \text{Não há deslocamento.}$$

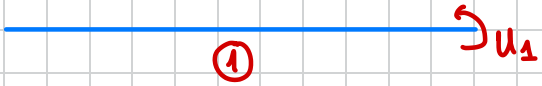
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(7)



1 g.l $\Rightarrow [K]_{1\times1}$

$L^{①} = 2L$



a) notação exata em B (x = 2L)

m.c

	1	2	3	4
①	-	-	-	1

$K_{11} = K_{44}^{①} = \frac{EI}{(2L)^3} \times 4(2L)^2 = 2\frac{EI}{L}$ //

$[K] = \frac{2EI}{L}$

$[F] = ?$

$f = P_2 \cdot \frac{x}{L} + P_1$

$\begin{cases} x = 0 \Rightarrow -\frac{P}{2} = P_2 \cdot \frac{0}{2L} + P_1 \Leftrightarrow P_1 = -\frac{P}{2} \text{ //} \\ x = 2L \Rightarrow -P = P_2 \cdot \frac{2L}{2L} + \left(-\frac{P}{2}\right) \Leftrightarrow P_2 = -\frac{P}{2} \text{ //} \end{cases}$

$\{F^{①}\} = \begin{Bmatrix} F_{1eq} \\ M_{1eq} \\ F_{2eq} \\ M_{2eq} \end{Bmatrix} \Rightarrow$ só há $M_2 \Rightarrow -\left(\frac{-\frac{P}{2} \times (2L)^2}{20}\right) - \left(\frac{-\frac{P}{2} \times (2L)^2}{12}\right) = \frac{4PL^2}{15}$ //

$[K][u] = [F] \Leftrightarrow \frac{2EI}{L} \times U_1 = \frac{4PL^2}{15} \Leftrightarrow U_1 = \frac{2PL^3}{15EI}$ //

b) deslocamento transversal aproximado em $x = L$

$v(x) = N_1 \cdot U_1 + N_2 \cdot U_2 + N_3 \cdot U_3 + N_4 \cdot U_4$

$v(x) = N_4 \, U_1$

$v(x) = \left(-\frac{x^2}{2L} + \frac{x^3}{(2L)^2}\right) \times \frac{2PL^3}{15EI}$

$v(x=L) = \left(-\frac{L^2}{2L} + \frac{L^3}{(2L)^2}\right) \times \frac{2PL^3}{15EI} = -\frac{PL^4}{30EI}$ //

c) Momento fletor ponto A ($x = 0$)

$M(x) = \frac{d^2v}{dx^2} \times EI$

$v(x) = \left(-\frac{x^2}{2L} + \frac{x^3}{(2L)^2}\right) \times \frac{2PL^3}{15EI}$

$$\cdot \frac{dv}{dx} = \left(-\frac{2x}{2L} + \frac{3x^2}{(2L)^2} \right) \times \frac{2pL^3}{15EI}$$

$$\cdot \frac{d^2v}{dx} = \left(-\frac{2}{2L} + \frac{6x}{(2L)^2} \right) \times \frac{2pL^3}{15EI} \quad //$$

$$\cdot M(x) = \left(-\frac{2}{2L} + \frac{6x}{(2L)^2} \right) \times \frac{2pL^3}{15Ex} \times \cancel{EI} = \left(-\frac{1}{L} + \frac{6x}{(2L)^2} \right) \times \frac{2pL^3}{15EI}$$

$$\cdot M(x=0) = -\frac{1}{L} \times 2pL^3 = -\frac{2pL^2}{15} \quad //$$

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⑧

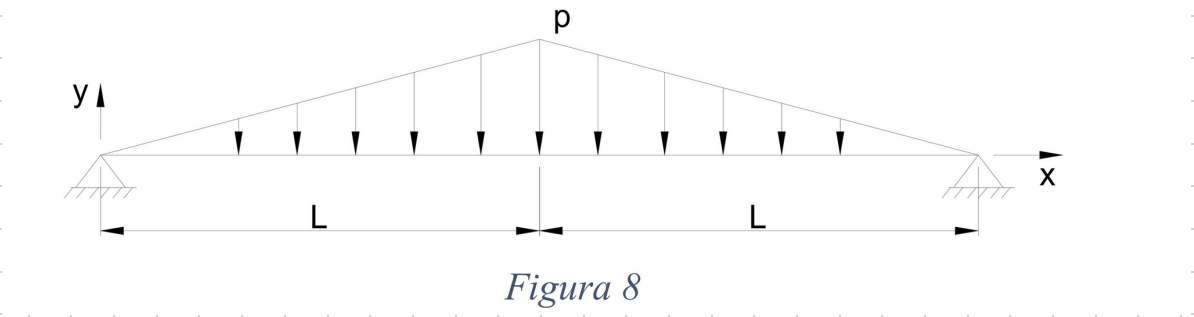
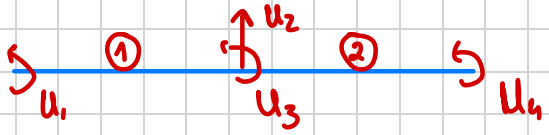


Figura 8

a) Deslocamento exato a meio da viga ($x = L$)



⇒ Aplicando simetria



2g.l ⇒ $[K]_{2\times2}$

$L^{①} = L$

m.c

	1	2	3	4	L
①	-	1	2	-	L

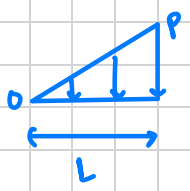
$$K_{11} = K_{22}^{①} = \frac{EI}{L^3} \times 4L^2 = \frac{4EI}{L} \quad //$$

$$K_{12} = K_{23}^{①} = \frac{EI}{L^3} \times (-6L) = -\frac{6EI}{L^2} \quad //$$

$$K_{12} = K_{33}^{①} = \frac{EI}{L^3} \times 12 = \frac{12EI}{L^3} \quad //$$

$$[K] = \frac{EI}{L^3} \begin{bmatrix} 4L^2 & -6L \\ -6L & 12 \end{bmatrix}$$

$$f = P_2 \cdot \frac{x}{L} + P_1$$



$$\begin{cases} x = 0 \Rightarrow 0 = P_2 \cdot \frac{0}{L} + P_1 \Leftrightarrow P_1 = 0 \\ x = L \Rightarrow -p = P_2 \cdot \frac{L}{L} + 0 \Leftrightarrow P_2 = -p \end{cases}$$

$$\{F\} = \begin{Bmatrix} F_{1eq} \\ M_{1eq} \\ F_{2eq} \\ M_{2eq} \end{Bmatrix}$$ Apenas temos M_{1eq} e F_{2eq}

$$M_{1eq} = \frac{-pL^2}{30} + \frac{0 \cdot L^2}{12} = \frac{-pL^2}{30} \quad //$$

$$F_{2eq} = \frac{7(-p)L}{20} + \frac{0L}{2} = -\frac{7pL}{20} \quad //$$

$$F = \begin{Bmatrix} -\frac{pL^2}{30} \\ -\frac{7pL}{20} \end{Bmatrix}$$

$$[K]\,[u] = [F] \Leftrightarrow \frac{EI}{L^3}\begin{bmatrix} 4L^2 & -6L \\ -6L & 12 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} -\frac{pL^2}{30} \\ -\frac{7pL}{20} \end{Bmatrix} \Leftrightarrow \begin{cases} \frac{4EI}{L} u_1 - 6\frac{EI}{L^2} u_2 = -\frac{pL^2}{30} \\ -6\frac{EI}{L^2} u_1 + 12\frac{EI}{L^3} u_2 = -\frac{7pL}{20} \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} u_1 = -\frac{5pL^3}{24EI} \quad // \\ u_2 = -\frac{2pL^4}{15EI} \quad // \end{cases} \Rightarrow \text{deslocamento exato a meio da viga}$$

b) Rotação aproximada em $L/2$.

$$\theta(x) = \frac{dv}{dx}$$

$$v(x) = N_1 \cdot u_1 + N_2 \cdot u_2 + N_3 \cdot u_3 + N_4 \cdot u_4$$

$$v(x) = N_2\, u_1 + N_3 \cdot u_2$$

$$v(x) = \left(x - \frac{2x^2}{L} + \frac{x^3}{L^2}\right) x\ \frac{-5pL^3}{24EI} + \left(\frac{3x^2}{L^2} - \frac{2x^3}{L^3}\right) x\ -\frac{2pL^4}{15EI}$$

$$\frac{dv}{dx} = \left(1 - \frac{4x}{L} + \frac{3x^2}{L^2}\right) x\ \frac{-5pL^3}{24EI} + \left(\frac{6x}{L^2} - \frac{6x^2}{L^3}\right) x\ \frac{-2pL^4}{15EI} \quad //$$

$$\theta(x) = \left(1 - \frac{4x}{L} + \frac{3x^2}{L^2}\right) x\ -\frac{5pL^3}{24EI} + \left(\frac{6x}{L^2} - \frac{6x^2}{L^3}\right) x\ \frac{-2pL^4}{15EI}$$

$$\theta(x=L/2) = \left(1 - \frac{4(L/2)}{L} + \frac{3(L/2)^2}{L^2}\right) \times \frac{-5pL^3}{24EI} + \left(\frac{6(L/2)}{L^2} - \frac{6(L/2)^2}{L^3}\right) \times \frac{-2pL^4}{15EI} \Leftrightarrow$$

$$\Leftrightarrow \theta(x=L/2) = \underbrace{\left(1 - \frac{4K}{2K} + \frac{3K^2}{4K^2}\right)}_{-1/4} \times \left(\frac{-5pL^3}{24EI}\right) + \underbrace{\left(\frac{6K}{2L^2} - \frac{6K^2}{4L^2}\right)}_{\frac{3}{L} - \frac{1,5}{L} = \frac{3}{2}L} \times \frac{-2pL^4}{15EI} \Leftrightarrow$$

$$\Leftrightarrow \theta(x=L/2) = \frac{-71pL^3}{480EI}$$

c) Esforço transverso em $x = \frac{3L}{2}$ $\Rightarrow$ Simetria $V(x=\frac{2L}{2}) = V(x=L/2)$

$$V(x) = \frac{dM}{dx} = d\left(\frac{d^2v}{dx}\right) \times EI$$

$$\cdot \frac{d^2v}{dx} = \left(\frac{4}{L} + \frac{6x}{L^2}\right) \times \frac{-5pL^3}{24EI} + \left(\frac{6}{L^2} - \frac{12x}{L^3}\right) \times \frac{-2pL^4}{15EI}$$

$$\cdot \frac{d^3v}{dx} = \left(\frac{6}{L^2}\right) \times \frac{-5pL^3}{24EI} + \left(-\frac{12}{L^3}\right) \times \left(\frac{-2pL^4}{15EI}\right)$$

$$V(x) = \frac{6}{L^2} \times \frac{-5pL^3}{24EI} + \left(-\frac{12}{L^3}\right) \times \left(\frac{-2pL^4}{15EI}\right) \times EI = \frac{6}{L^2} \times \frac{-5pL^3}{24} + \left(-\frac{12}{L^3}\right) \times \left(\frac{-2pL^4}{15}\right) =$$

$$V(x) = \frac{7pL}{20}$$

$$V(x=L/2) = V(x) = \frac{7pL}{20} //$$

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⑨

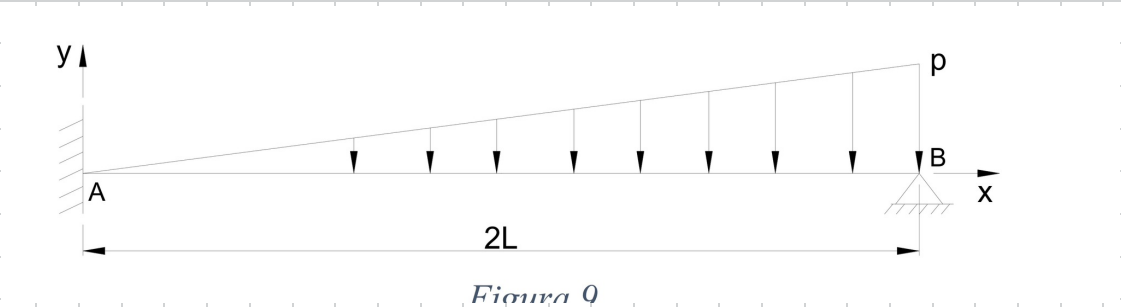
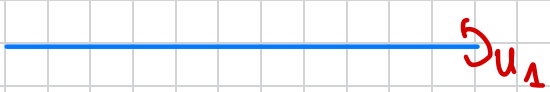


Figura 0



$\Rightarrow$ Como queremos reacção no ponto B acrescentamos + 1 grau

1 g l $\Rightarrow$ $[K]_{1\times1}$



2 g l $\Rightarrow$ $[K]_{2\times2}$

$L^{①} = 2L$

m.c

	1	2	3	4
①	-	-	1	2

onde $U_1 = 0$

$$K_{11} = K_{33}^{①} = \frac{EI}{(2L)^3} \times 12 = \frac{3EI}{2L^3}$$

$$K_{12} = K_{34}^{①} = \frac{EI}{(2L)^3} \times -6(2L) = -\frac{3EI}{2L^2}$$

$$K_{22} = K_{44}^{①} = \frac{EI}{(2L)^3} \times 4(2L)^2 = \frac{2EI}{L}$$

$$[K] = \frac{EI}{L^3} \begin{bmatrix} 3/2 & -3/2L \\ -3/2L & 2L^2 \end{bmatrix}$$

$$\{F\} = \begin{Bmatrix} F_{1eq} \\ M_{1eq} \\ F_{2eq} \\ M_{2eq} \end{Bmatrix}$$ } Apenas temos F_{2eq} e M_{2eq}

$$f = P_2 \cdot \frac{x}{L} + P_1$$



$$\begin{cases} x = 0 \Rightarrow 0 = P_2 \cdot \frac{0}{2L} + P_1 \iff P_1 = 0 \\ x = 2L \Rightarrow -p = P_2 \cdot \frac{2L}{2L} + 0 \iff P_2 = -p \end{cases}$$

- $F_{2eq} = \frac{7(-p) \times 2L}{20} + \frac{0 \times 2L}{2} = -\frac{14pL}{20} = -\frac{7pL}{10}$

- $M_{2eq} = -\left(\frac{-p \times (2L)^2}{20} - \frac{0 \times (2L)^2}{12} \right) = \frac{4pL^2}{20} = \frac{pL^2}{5}$

$$\{F\} = \begin{Bmatrix} -\frac{7pL}{10} \\ \frac{pL^2}{5} \end{Bmatrix}$$

$$[K]\,[u] = [F]$$

$$\frac{EI}{L^3} \begin{bmatrix} 3/2 & -3/2L \\ -3/2L & 2L^2 \end{bmatrix} \begin{Bmatrix} U_1 \overset{0}{\not{}} \\ U_2 \end{Bmatrix} = \begin{Bmatrix} -\frac{7pL}{10} + R_B \\ \frac{pL^2}{5} \end{Bmatrix} \iff \begin{cases} \frac{3}{2}\frac{EI}{L^3} \cdot \not{U_1}^{0} - \frac{3}{2}\frac{EI}{L^2} \cdot U_2 = -\frac{7pL}{10} + R_B \\ -\frac{3}{2}\frac{EI}{L^2} \times \not{U_1}^{0} + \frac{2EI}{L} U_2 = \frac{pL^2}{5} \end{cases}$$

$$\Leftrightarrow \begin{cases} -\frac{3}{2}\frac{EI}{L^2}\cdot U_2 = -\frac{7pL}{10} + R_B \\ \frac{2EI}{L}\cdot U_2 = \frac{pL^2}{5} \end{cases} \Leftrightarrow \begin{cases} U_1 = 0 \\ U_2 = \frac{pL^3}{10EI} \\ R_B = \frac{11pL}{20} \end{cases}$$

$U_2 = \frac{pL^3}{10EI}$ // ⇒ Rotação em B // a)

$R_B = \frac{11pL}{20}$ // ⇒ Reação em B // c)

b) $\theta(x) = 0$

$\theta(x) = \frac{dv}{dx}$

$v(x) = N_1 \cdot U_1 + N_2 \cdot U_2 + N_3 \cdot U_3 + N_4 \cdot U_4$

$v(x)_{real} = N_4 \cdot U_1 = \left(-\frac{x^2}{2L} + \frac{x^3}{(2L)^2}\right) \times \frac{pL^3}{10EI}$

do 1: diagrama

$\cdot \frac{dv}{dx} = \left(-\frac{2x}{2L} + \frac{3x^2}{(2L)^2}\right) \times \frac{pL^3}{10EI}$

$\cdot \theta(x) = \left(-\frac{2x}{2L} + \frac{3x^2}{(2L)^2}\right) \times \frac{pL^3}{10EI}$

$\cdot \theta(x) = 0 \Leftrightarrow \left(-\frac{2x}{2L} + \frac{3x^2}{(2L)^2}\right) \times \frac{pL^3}{10EI} = 0 \Leftrightarrow -\frac{2pL^2x}{20EI} + \frac{3x^2pL^3}{40EIL^2} = 0 \Leftrightarrow$

$\Leftrightarrow -\frac{pLx^2}{10} + \frac{3x^2pL}{40} = 0 \Leftrightarrow x = 0$ // $\vee$ $x = \frac{4L}{3}$ //

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(10)

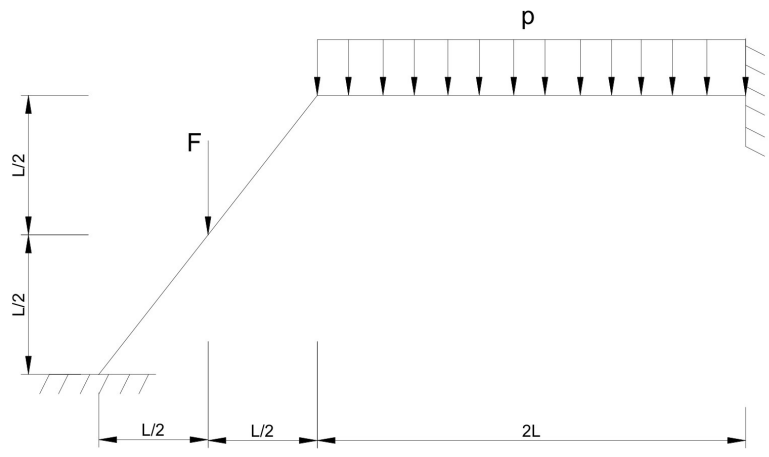
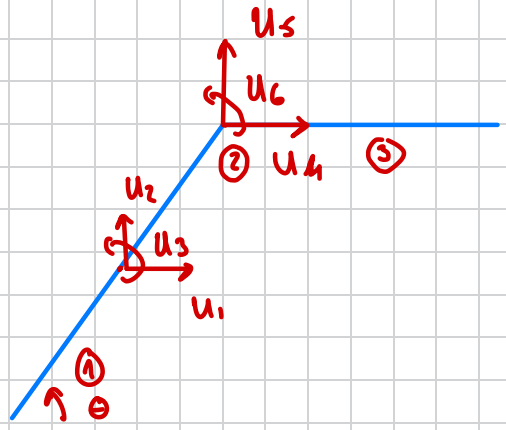


Figura 10



$L^{①} = \sqrt{(L/2)^2 + (L/2)^2} = 0{,}707L$

$L^{②} = L^{①} = 0{,}707L$

$L^{③} = 2L$

6g.l ⇒ $[K]_{6\times6}$

$\theta = tg^{-1}\left(\frac{L/2}{L/2}\right) = tg^{-1}(1) = 45°$

m.c

	1	2	3	4	5	6	θ	L
①	-	-	-	1	2	3	45°	0,707L
②	1	2	3	4	5	6	45°	0,707L
③	4	5	6	-	-	-	0	2L

$$K_{11} = K_{44}^{①} + K_{11}^{②} = \frac{\cos^2 45\, EA}{0,707L} + \frac{12\sin^2(45)\, EI}{(0,707L)^3} + \frac{\cos^2(45)\, EA}{0,707L} + \frac{12\sin^2(45) \times EI}{(0,707L)^3} =$$

$$= \frac{EA}{L} \times 1,414 + \frac{EI}{L^3} \times 33,96 //$$

$$K_{12} = K_{45}^{①} + K_{12}^{②} = \frac{\sin 45 \cos 45\, EA}{0,707L} - \frac{12\sin 45 \cos 45\, EI}{(0,707L)^3} + \frac{\sin 45 \cos 45\, EA}{0,707L} - \frac{12\sin(45)\cos(45)\, EI}{(0,707L)^3} =$$

$$= \frac{EA}{L} \times 1,414 + \frac{EI}{L^3} \times (-33,96) //$$

$$K_{13} = K_{46}^{①} + K_{13}^{②} = \frac{6.\sin(45).EI}{(0,707L)^2} - \frac{6\sin(45)\, EI}{(0,707L)^2} = 0 //$$

$$K_{14} = K_{14}^{②} = -\frac{\cos^2(45) \times EA}{0,707L} - \frac{12\sin^2 45 \times EI}{(0,707L)^3} = \frac{EA}{L} \times (-0,707) + \frac{EI}{L^3}(-16,98)$$

$$K_{15} = K_{15}^{②} = -\frac{\sin 45 \cos 45\, EA}{0,707L} + \frac{12\sin 45 \cos 45\, EI}{(0,707L)^3} = \frac{EA}{L} \times (-0,707) + \frac{EI}{L^3}(16,98) //$$

$$K_{16} = K_{16}^{②} = -\frac{6\sin(45)\, EI}{(0,707L)^2} = \frac{EI}{L^2} \times (-8,488) //$$

$$K_{22} = K_{55}^{①} + K_{22}^{②} = \frac{\sin^2 45\, EA}{0,707L} + \frac{12\cos^2(45)\, EI}{(0,707L)^3} + \frac{\sin^2(45)\, EA}{0,707L} + \frac{12\cos^2 45\, EI}{(0,707L)^3} = \frac{EA}{L} \times 1,414 + \frac{EI}{L^3} \times 33,96 //$$

$$K_{23} = K_{56}^{①} + K_{23}^{②} = -\frac{6\cos 45\, EI}{(0,707L)^2} + \frac{6\cos(45)\, EI}{(0,707L)^2} = 0 //$$

$$K_{24} = K_{24}^{②} = -\frac{\sin 45 \cos 45\, EA}{0,707L} + \frac{12\sin 45 \cos 45\, EI}{(0,707L)^3} = \frac{EA}{L} \times (-0,707) + \frac{EI}{L^3} \times (16,98) //$$

$$K_{25} = K_{25}^{②} = -\frac{\sin^2(45)\, EA}{0,707L} - \frac{12\cos^2(45)\, EI}{(0,707L)^3} = \frac{EA}{L} \times (-0,707) + \frac{EI}{L^3} \times (-16,98) //$$

$$K_{26} = K_{26}^{②} = \frac{6\cos(45) \times EI}{(0,707L)^2} = \frac{EI}{L^2} \times (8,488) //$$

$$K_{33} = K_{66}^{①} + K_{33}^{②} = \frac{4EI}{0,707L} + \frac{4EI}{0,707L} = \frac{EI}{L} \times 11,315 //$$

$$K_{34} = K_{34}^{②} = \frac{6\sin 45 \times EI}{(0,707L)^2} = \frac{EI}{L^2} \times (8,488) //$$

$$K_{35} = K_{35}^{②} = -\frac{6\cos 45 \times EI}{(0{,}707L)^2} = \frac{EI}{L^2} \times (-8{,}488)$$

$$K_{36} = K_{36}^{②} = \frac{2EI}{0{,}707L} = \frac{EI}{L} \times 2{,}829 //$$

$$K_{44} = K_{44}^{②} + K_{11}^{①} = \frac{\cos^2 45\, EA}{0{,}707L} + \frac{12\sin^2(45)\, EI}{(0{,}707L)^3} + \frac{\cos^2(0)\, EA}{2L} + \frac{12\sin^2(0)\, EI}{(2L)^2} = \frac{EA}{L} \times 1{,}21 + \frac{EI}{L^3} \times 16{,}98 //$$

$$K_{45} = K_{45}^{②} + K_{12}^{③} = \frac{\sin 45 \cos 45\, EA}{0{,}707L} - \frac{12\sin 45 \cos 45\, EI}{(0{,}707L)^3} + \frac{\sin(0)\cos(0)\, EA}{2L} - \frac{12\sin 0 \cos 0\, EI}{(2L)^3} = \frac{EA}{L} \times 0{,}707 + \frac{EI}{L^3} \times (-16{,}98) //$$

$$K_{46} = K_{46}^{②} + K_{13}^{③} = \frac{6\sin 45\, EI}{(0{,}707L)^2} - \frac{6\sin(0)EI}{(2L)^2} = \frac{EI}{L^2} \times (8{,}488) //$$

$$K_{55} = K_{55}^{②} + K_{22}^{①} = \frac{\sin^2(45)\, EA}{0{,}707L} + \frac{12\cos^2 45\, EI}{(0{,}707L)^3} + \frac{\sin^2(0)\, EA}{2L} + \frac{12\cos^2 0\, EI}{(2L)^3} = \frac{EA}{L} \times 0{,}707 + \frac{EI}{L^3} \times 18{,}48 //$$

$$K_{56} = K_{56}^{②} + K_{23}^{③} = -\frac{6\cos 45\, EI}{(0{,}707L)^2} + \frac{6\cos(0)\, EI}{(2L)^2} = \frac{EI}{L^2} \times (-6{,}99) //$$

$$K_{66} = K_{66}^{②} + K_{33}^{③} = \frac{4EI}{0{,}707L} + \frac{4EI}{2L} = \frac{EI}{L} \times 7{,}658 //$$

$$[K] = \begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} & K_{15} & K_{16} \\ & K_{22} & K_{23} & K_{24} & K_{25} & K_{26} \\ & & K_{33} & K_{34} & K_{35} & K_{36} \\ & S & & K_{44} & K_{45} & K_{46} \\ & & & & K_{55} & K_{56} \\ & & & & & K_{66} \end{bmatrix}$$

b) Elemento 1. $f^{①} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -F \\ 0 \end{Bmatrix}$ ⇒ só se usa no fim

Elemento 2: $f^{②} = \begin{Bmatrix} 0 \\ -F \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$ ⇒ só se usa no fim

Elemento 3: $\begin{Bmatrix} 0 \\ -\frac{p \times 2L}{2} \\ -\frac{p \times (2L)^2}{12} \\ 0 \\ 0 \\ 0 \end{Bmatrix} = f^{③} \iff f^{③} = \begin{Bmatrix} 0 \\ -pL \\ -\frac{pL^2}{3} \\ 0 \\ 0 \\ 0 \end{Bmatrix}$

Vetor de forças:

$$f_1 = f_4^{①} + f_1^{②} = 0$$
$$f_2 = f_5^{①} + f_2^{②} = 0$$
$$f_3 = f_6^{①} + f_3^{②} = 0$$
$$f_4 = f_4^{②} + f_1^{③} = 0$$
$$f_5 = f_5^{②} + f_2^{③} = 0 + (-pL) = -pL$$
$$f_6 = f_6^{②} + f_3^{③} = 0 + \left(-\frac{pL^2}{3}\right) = -\frac{pL^2}{3}$$

$$\Rightarrow \{F\} = \begin{Bmatrix} 0 \\ -F \\ 0 \\ 0 \\ -pL \\ -\frac{pL^2}{3} \end{Bmatrix}$$

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(11)

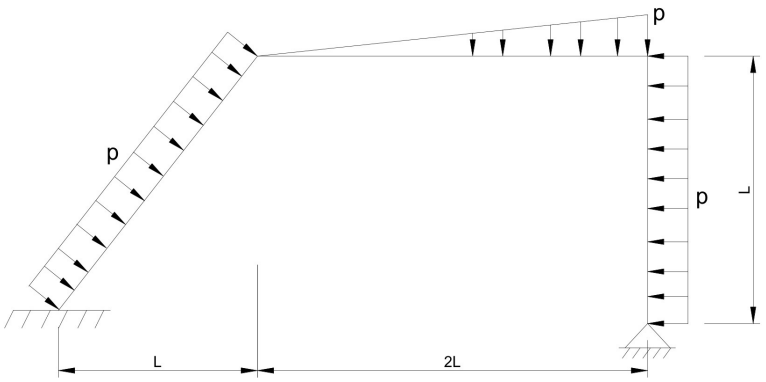
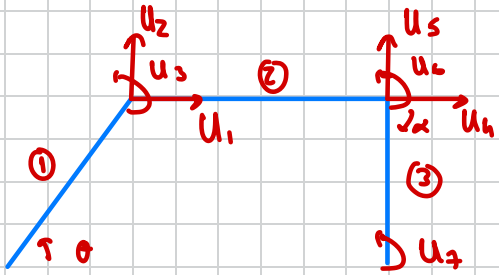


Figura 11



7 g.l ⇒ $[K]_{7\times7}$

$$L^{①} = \sqrt{L^2 + L^2} = 1{,}414L$$
$$L^{②} = 2L$$
$$L^{③} = L$$
$$\theta = tg^{-1}\left(\frac{L}{L}\right) = tg^{-1}(1) = 45^{\circ}$$

m.c

	1	2	3	4	5	6	θ	L
①	-	-	-	1	2	3	45	1,414L
②	1	2	3	4	5	6	0	2L
③	4	5	6	-	-	7	-90°	L

$$K_{11} = K_{44}^{①} + K_{11}^{②} = \frac{\cos^2 45 \times EA}{1{,}414L} + \frac{12\sin^2(45)EI}{(1{,}414L)^3} + \frac{\cos^2(0) \times EA}{2L} + \frac{12\sin^2(0) \times EI}{(2L)^3} = \frac{EA}{L} \times 0{,}853 + \frac{EI}{L^3} \times 2{,}12$$

$$K_{12} = K_{45}^{①} + K_{12}^{②} = \frac{\sin 45\cos 45\, EA}{1{,}414L} - \frac{12\sin(45)\cos(45)EI}{(1{,}414L)^3} + \frac{\sin 0 \cos 0\, EA}{2L} - \frac{12\sin 0\cos 0\, EI}{(2L)^3} =$$

$$= \frac{EA}{L} \times 0{,}353 + \frac{EI}{L^3} \times (-2{,}12)$$

$$K_{13} = K_{46}^{①} + K_{13}^{②} = \frac{6\sin 45^{\circ} EI}{(1{,}414L)^2} - \frac{6\sin(0)EI}{(2L)^2} = \frac{EI}{L^2} \times 2{,}12$$

$$K_{14} = K_{14}^{②} = -\frac{\cos^2(0)EA}{2L} - \frac{12\sin^2(0)EI}{(2L)^3} = \frac{EA}{L} \times (-0{,}5)$$

$$K_{15} = K_{15}^{②} = -\frac{\sin 0 \cos 0\, EA}{2L} + \frac{12\sin 0\cos 0 EI}{(2L)^3} = 0$$

$$K_{16} = K_{16}^{(2)} = -\frac{6\sin 0^{\circ} EI}{(2L)^2} = 0$$

$$K_{17} = 0$$

$$K_{22} = K_{55}^{(1)} + K_{22}^{(2)} = \frac{\sin^2 45 EA}{1{,}414L} + \frac{12\cos^2 45 EI}{(1{,}414L)^3} + \frac{\sin^2 0^{\circ} EA}{2L} + \frac{12\cos^2 0 EI}{(2L)^3} = \frac{EA}{L} \times 0{,}353 + \frac{EI}{L^3} \times 3{,}62$$

$$K_{23} = K_{56}^{(1)} + K_{23}^{(2)} = -\frac{6\cos 45 EI}{(1{,}414L)^2} + \frac{6\cos 0 EI}{(2L)^2} = \frac{EI}{L^2} \times (-0{,}62)$$

$$K_{24} = K_{24}^{(1)} = -\frac{\sin(0)\cos(0) EA}{2L} + \frac{12\sin 0\cos 0 EI}{(2L)^3} = 0$$

$$K_{25} = K_{25}^{(2)} = -\frac{\sin^2 0^{\circ} EA}{2L} - \frac{12\cos^2 0 EI}{(2L)^3} = \frac{EI}{L^3} \times \left(-\frac{3}{2}\right)$$

$$K_{26} = K_{26}^{(2)} = \frac{6\cos 0 EI}{(2L)^2} = \frac{EI}{L^2} \times \frac{3}{2}$$

$$K_{27} = 0$$

$$K_{33} = K_{66}^{(1)} + K_{33}^{(2)} = \frac{4EI}{1{,}414L} + \frac{4EI}{2L} = \frac{EI}{L} \times 4{,}83$$

$$K_{34} = K_{34}^{(2)} = \frac{6 \times \sin 0^{\circ} \times EI}{(2L)^2} = 0$$

$$K_{35} = K_{35}^{(2)} = -\frac{6\cos 0 \times EI}{(2L)^2} = \frac{EI}{L^2} \times \left(-\frac{3}{2}\right)$$

$$K_{36} = K_{36}^{(2)} = \frac{2EI}{2L} = \frac{EI}{L}$$

$$K_{37} = 0$$

$$K_{44} = K_{44}^{(2)} + K_{11}^{(3)} = \frac{\cos^2 0 EA}{2L} + \frac{12\sin^2(0) EI}{(2L)^3} + \frac{\cos^2(-90^{\circ}) EA}{L} + \frac{12\sin^2(-90) EI}{L^3} = \frac{EA}{L} \times \frac{1}{2} + \frac{EI}{L^3} \times 12$$

$$K_{45} = K_{45}^{(2)} + K_{12}^{(3)} = \frac{\sin 0^{\circ}\cos 0 EA}{2L} - \frac{12\sin 0^{\circ}\cos 0 EI}{(2L)^3} + \frac{\sin(-90)\cos(-90^{\circ}) EA}{L} - \frac{12\sin -90\cos^{0} -90 EI}{L^3} = 0$$

$$K_{46} = K_{46}^{(2)} + K_{13}^{(3)} = \frac{6\sin 0^{\circ} EI}{(2L)^2} - \frac{6\sin(-90) EI}{L^2} = \frac{EI}{L^2} \times 6$$

$$K_{47} = K_{16}^{(3)} = -\frac{6\sin(-90) EI}{L^2} = \frac{EI}{L^2} \times 6$$

$$K_{55} = K_{55}^{(1)} + K_{22}^{(3)} = \frac{\sin^2 0^{\circ} EA}{2L} + \frac{12\cos^2 0 EI}{(2L)^3} + \frac{\sin^2(-90) EA}{L} + \frac{12\cos^2 -90 EI}{L^3} = \frac{EA}{L} \times (-1) + \frac{EI}{L^3} \times \frac{3}{2}$$

$$K_{56} = K_{56}^{(2)} + K_{23}^{(3)} = -\frac{6\cos 0 \times EI}{(2L)^2} + \frac{6\cos(90^\circ) EI}{L^2} = \frac{EI}{L^2} \times \left(-\frac{3}{2}\right) //$$

$$K_{57} = K_{26}^{(3)} = \frac{6\cos(-90^\circ) \times EI}{L^2} = 0 //$$

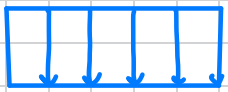
$$K_{66} = K_{66}^{(2)} + K_{33}^{(3)} = \frac{4EI}{2L} + \frac{4EI}{L} = \frac{EI}{L} \times 6 //$$

$$K_{67} = K_{36}^{(3)} = \frac{2EI}{L} = \frac{EI}{L} \times 2 //$$

$$K_{77} = K_{66}^{(3)} = \frac{4EI}{L} = \frac{EI}{L} \times 4 //$$

b) Vetor de forças

Elemento 1:



$$\{f\}^{(1)} = \left\{ 0 ; \frac{-p \times 1,414L}{2} ; \frac{-p \times (1,414L)^2}{12} ; 0 ; \frac{-p \times 1,414L}{2} ; \frac{-(-p) \times (1,414L)}{12} \right\}^T$$

$$\{f\}^{(1)} = \begin{Bmatrix} 0 \\ -0,707\,pL \\ -0,167\,pL^2 \\ 0 \\ -0,707\,pL \\ 0,167\,pL^2 \end{Bmatrix}$$

está inclinado ⇒ $\{f\} = \{R^T\}\{F\}$

⇓ transpôr

$$[F]^{(1)} = \begin{bmatrix} \cos 45 & -\sin 45 & 0 & 0 & 0 & 0 \\ \sin 45 & \cos 45 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos 45 & -\sin 45 & 0 \\ 0 & 0 & 0 & \sin 45 & \cos 45 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ -0,707\,pL \\ -0,167\,pL^2 \\ 0 \\ -0,707\,pL \\ 0,167\,pL^2 \end{Bmatrix} \Leftrightarrow [F]^{(1)} = \begin{Bmatrix} 0,5\,pL \\ -0,5\,pL \\ -0,167\,pL^2 \\ 0,5\,pL \\ -0,5\,pL \\ 0,167\,pL^2 \end{Bmatrix}$$

Só há estes

Elemento 2



$$f = P_2 \cdot \frac{x}{L} + P_1$$

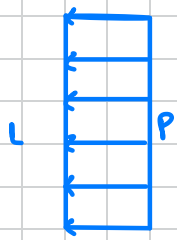
$x = 0 \Rightarrow 0 = P_2 \cdot \frac{0}{2L} + P_1 \Leftrightarrow P_1 = 0$

$x = 2L \Rightarrow -p = P_2 \cdot \frac{2L}{2L} + 0 \Leftrightarrow P_2 = -p$

$$\{F^{(2)}\} = \begin{Bmatrix} 0 \\ \frac{3(-p) \times 2L}{20} + \frac{0 \times 2L}{2} \\ \frac{-p \times (2L)^2}{30} + \frac{0 \times (2L)^2}{12} \\ 0 \\ \frac{3(-p) \times (2L)}{20} + \frac{0 \times L}{2} \\ -\frac{(-p) \times (2L)^2}{20} - \frac{0 \times L^2}{12} \end{Bmatrix} = \begin{Bmatrix} 0 \\ -0,3\,pL \\ -0,133\,pL^2 \\ 0 \\ -0,7\,pL \\ 0,2\,pL^2 \end{Bmatrix}$$

Temos todos

Elemento 3



De modo a não transpor consideramos o vetor de forças diretamente inclinado visto ser 90°

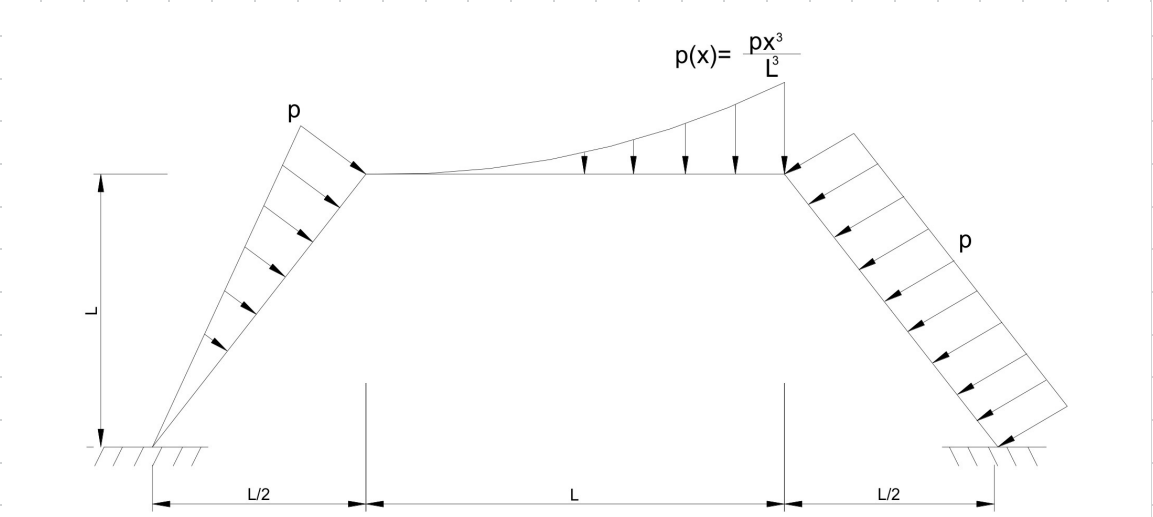
$$\{F^{①}\} = \begin{Bmatrix} -\frac{pL}{2} \\ 0 \\ -\frac{pL^2}{12} \\ -\frac{pL}{2} \\ 0 \\ -\frac{(-p)L^2}{12} \end{Bmatrix} = \begin{Bmatrix} -0{,}5pL \\ 0 \\ -0{,}0833\,pL^2 \\ -0{,}5\,pL \\ 0 \\ +0{,}0833\,pL^2 \end{Bmatrix} \begin{matrix} \} \text{Temos os 3} \\ \\ \\ \\ \} \text{Temos} \end{matrix}$$

Vetor de forças geral:

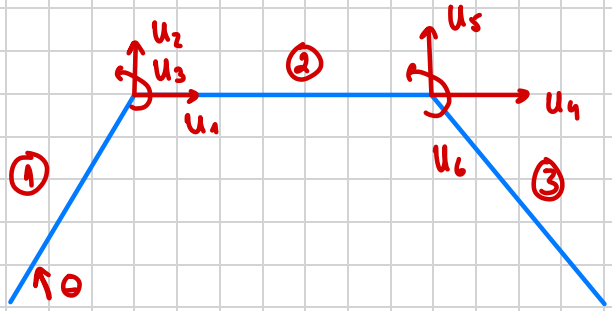
$$\{F\} = \begin{Bmatrix} 0{,}5pL + 0 \\ -0{,}5pL - 0{,}3pL \\ 0{,}167\,pL^2 - 0{,}133\,pL^2 \\ 0 - 0{,}5pL \\ -0{,}7\,pL + 0 \\ 0{,}2pL^2 - 0{,}0833pL^2 \\ +0{,}0833\,pL^2 \end{Bmatrix} = \begin{Bmatrix} 0{,}5pL \\ -0{,}8pL \\ 0{,}034\,pL^2 \\ -0{,}5pL \\ -0{,}7pL \\ 0{,}1167\,pL^2 \\ 0{,}0833\,pL^2 \end{Bmatrix} //$$

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⑫



a) Vetor de forças



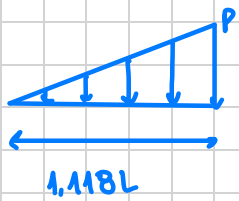
$$L^{①} = \sqrt{L^2 + (L/2)^2} = 1{,}118L$$

$$L^{②} = L$$

$$L^{③} = L^{①} = 1{,}118L$$

$$\theta = tg^{-1}\left(\frac{L}{L/2}\right) = 63{,}435°$$

Elemento 1



$$f = P_2 \cdot \frac{x}{L} + P_1 \Rightarrow \begin{cases} x = 0 \Rightarrow 0 = P_2 \times \frac{0}{1,118L} + P_1 \Leftrightarrow P_1 = 0 \\ x = 1,118L \Rightarrow -p = P_2 \times \frac{1,118L}{1,118L} + 0 \Leftrightarrow P_2 = -p \end{cases}$$

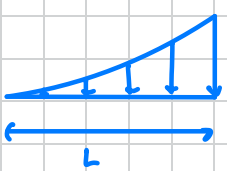
$$f^{①} = \left\{ 0 \; ; \; \frac{3(-p) \times 1,118L}{20} \; ; \; \frac{-p \times (1,118L)^2}{30} \; ; \; 0 \; ; \; \frac{7(-p) \times 1,118L}{20} \; ; \; -\frac{(-p) \times (1,118L)^2}{20} \right\}^T$$

$$f^{①} = \begin{Bmatrix} 0 \\ -0,1677\,pL \\ -0,04167\,pL^2 \\ 0 \\ -0,3913\,pL \\ 0,0625\,pL^2 \end{Bmatrix} \qquad \{F^{①}\} = [R]^T \times [f]$$

$$\{F\} = \begin{bmatrix} \cos 63,435 & -\sin 63,435 & 0 & 0 & 0 & 0 \\ \sin 63,435 & \cos 63,435 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos 63,435 & -\sin 63,435 & 0 \\ 0 & 0 & 0 & \sin 63,435 & \cos 63,435 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ -0,1677\,pL \\ -0,04167\,pL^2 \\ 0 \\ -0,3913\,pL \\ 0,0625\,pL^2 \end{Bmatrix}$$

$$\Leftrightarrow \{F^{①}\} = \begin{Bmatrix} 0,15\,pL \\ -0,075\,pL \\ -0,04167\,pL^2 \\ 0,35\,pL \\ -0,175\,pL \\ 0,0625\,pL^2 \end{Bmatrix}$$ Usar apenas estes

Elemento ②



$$\{F^{②}\} = \{0 \; ; \; F_2^{①} \; ; \; F_3^{①} \; ; \; 0 \; , \; F_5^{①} \; ; \; F_6^{②}\}$$

$$F_2^{②} = -\int_0^L p(x) N_1 \, dx = -\int_0^L \left(\frac{px^3}{L^3}\right) \cdot \left(1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}\right) dx = -\int_0^L \frac{px^3}{L^3} - \frac{3px^5}{L^5} + \frac{2px^6}{L^6} dx = -0,036\,pL$$

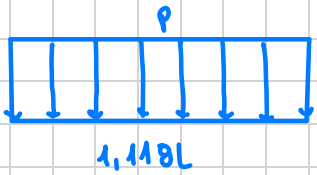
$$F_3^{②} = -\int_0^L p(x) N_2 \, dx = -\int_0^L \left(\frac{px^3}{L^3}\right) \cdot \left(x - \frac{2x^2}{L} + \frac{x^3}{L^2}\right) dx = -\int_0^L \frac{px^4}{L^3} - \frac{2px^5}{L^4} + \frac{px^6}{L^5} dx = -0,0095\,pL^2$$

$$F_5^{②} = -\int_0^L p(x) N_3 \, dx = -\int_0^L \left(\frac{px^3}{L^3}\right) \cdot \left(\frac{3x^2}{L^2} - \frac{2x^3}{L^3}\right) dx = -\int_0^L \frac{3px^5}{L^5} - \frac{2px^6}{L^6} dx = -0,214\,pL$$

$$F_6^{②} = -\int_0^L p(x) N_4 \, dx = -\int_0^L \left(\frac{px^3}{L^3}\right) \cdot \left(-\frac{x^2}{L} + \frac{x^3}{L^2}\right) dx = -\int_0^L -\frac{px^5}{L^4} + \frac{px^6}{L^5} \, dx = 0{,}024\ pL^2$$

$$F^{②} = \left\{ \begin{array}{c} 0 \\ -0{,}036\ pL \\ -0{,}0035\ pL^2 \\ 0 \\ -0{,}214\ pL \\ 0{,}024\ pL^2 \end{array} \right\} \quad \text{Usar todos}$$

Elemento ③



$$\{g^{③}\} = \left\{ 0;\ -\frac{p \times 1{,}118L}{2};\ -\frac{p \times (1{,}118L)^2}{12};\ 0;\ -\frac{p \times 1{,}118L}{2};\ -\frac{(-p) \times (1{,}118L)^2}{12} \right\}^T$$

$$\{g^{③}\} = \left\{ \begin{array}{c} 0 \\ -0{,}559\ pL \\ -0{,}104\ pL^2 \\ 0 \\ -0{,}559\ pL \\ 0{,}104\ pL^2 \end{array} \right\} \qquad \{F^{③}\} = [R^T]\{g\}$$

$$\{F^{③}\} = \begin{bmatrix} \cos -63{,}435 & -\sin -63{,}435 & 0 & 0 & 0 & 0 \\ \sin 63{,}435 & \cos \cdot 63{,}435 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos -63{,}435 & -\sin \cdot 63{,}435 & 0 \\ 0 & 0 & 0 & \sin \cdot 63{,}435 & \cos -63{,}435 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \left\{ \begin{array}{c} 0 \\ -0{,}559 pL \\ -0{,}104 pL^2 \\ 0 \\ -0{,}559 pL \\ 0.104 pL^2 \end{array} \right\}$$

$$[F^{③}] = \left\{ \begin{array}{c} -0{,}5\ pL \\ -0{,}249\ pL \\ -0{,}104\ pL^2 \\ -0{,}5 pL \\ -0{,}249\ pL \\ 0{,}104\ pL^2 \end{array} \right\} \quad \text{Usar Apenas Estes}$$

Vetor de forças:

$$[F] = \left\{ \begin{array}{c} 0{,}35\ pL + 0 \\ -0{,}135\ pL - 0{,}1036 pL \\ 0{,}0625\ pL^2 - 0{,}0035\ pL^2 \\ 0 - 0{,}5 pL \\ -0{,}214 pL - 0{,}249 pL \\ 0{,}024 pL^2 + 0{,}104\ pL^2 \end{array} \right\} \Leftrightarrow [F] = \left\{ \begin{array}{c} 0{,}35 pL \\ -0{,}211 pL \\ 0{,}053 pL^2 \\ -0{,}5 pL \\ -0{,}463\ pL \\ 0{,}128 pL^2 \end{array} \right\}$$

//

13

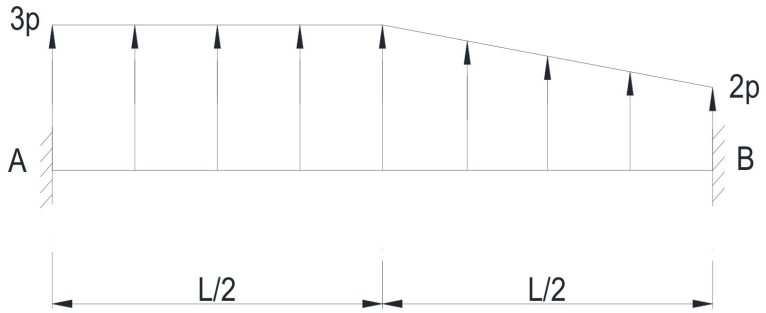
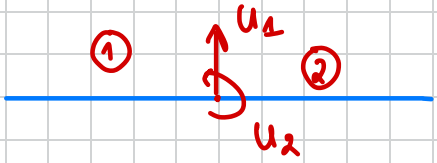


Figura 13



2g.l $\Rightarrow [K]_{2\times2}$

$L^{①} = L/2$

$L^{②} = L/2$

a) Rotação exata a meio da viga

m.c

	1	2	3	4	L
①	-	-	1	2	L/2
②	1	2	-	-	L/2

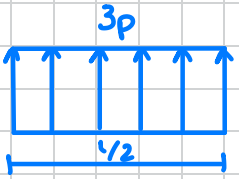
$K_{11} = K_{33}^{①} + K_{11}^{②} = \frac{EI}{(L/2)^3} \times 12 + \frac{EI}{(L/2)^3} \times 12 = 192\frac{EI}{L^3}$ //

$K_{12} = K_{34}^{①} + K_{12}^{②} = \frac{EI}{(L/2)^3} \times -6(L/2) + \frac{EI}{(L/2)^3} \times 6\times(L/2) = 0$ //

$K_{22} = K_{44}^{①} + K_{22}^{②} = \frac{EI}{(L/2)^3} \times 4\times(L/2)^2 + \frac{EI}{(L/2)^3} \times 4\times(L/2)^2 = 16\frac{EI}{L}$

$$[K] = \frac{EI}{L^3}\begin{bmatrix} 192 & 0 \\ 0 & 16L^2 \end{bmatrix}$$

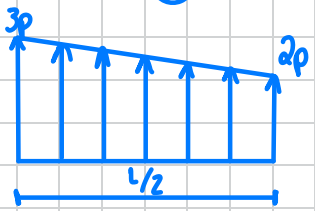
Elemento ①:



$$\{F^{①}\} = \left\{0;\ 0;\ \frac{3p\times(L/2)}{2};\ -\frac{3p\times(L/2)^2}{12}\right\}^T = \{0;\ 0;\ 0{,}75pL;\ -0{,}0625\,pL^2\}^T$$

$$F^{①} = \begin{Bmatrix} 0 \\ 0 \\ 0{,}75pL \\ -0{,}0625pL^2 \end{Bmatrix}$$

Elemento ②



$$f = P_2 \cdot \frac{x}{L} + P_1 \Rightarrow \begin{cases} x = 0 \Rightarrow 3p = P_2 \cdot \frac{0}{L/2} + P_1 \Leftrightarrow P_1 = 3p \\ x = L/2 \Rightarrow 2p = P_2 \cdot \frac{L/2}{L/2} + 3p \Leftrightarrow P_2 = -p \end{cases}$$

$$\{F^{①}\} = \left\{ \frac{3(-p)\times(L/2)}{20} + \frac{3p\times(L/2)}{2} \;;\; \frac{-p\times(L/2)^2}{30} + \frac{3p\times(L/2)^2}{12} \;;\; 0 \;;\; 0 \right\}^T$$

$$\{F^{②}\} = \begin{Bmatrix} 0{,}675\,pL \\ 0{,}0542\,pL^2 \\ 0 \\ 0 \end{Bmatrix}$$

$$\{F\} = \begin{Bmatrix} 0{,}75\,pL + 0{,}675\,pL \\ -0{,}0625\,pL^2 + 0{,}0542\,pL^2 \end{Bmatrix} = \begin{Bmatrix} 1{,}425\,pL \\ -0{,}0083\,pL^2 \end{Bmatrix}$$

$$[K]\,[u] = [F] \Leftrightarrow \frac{EI}{L^3}\begin{bmatrix} 192 & 0 \\ 0 & 16L^2 \end{bmatrix}\begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 1{,}425\,pL \\ -0{,}0083\,pL^2 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 192\frac{EI}{L^3}\cdot u_1 = 1{,}425\,pL \\ 16\frac{EI}{L}\cdot u_2 = -0{,}0083\,pL^2 \end{cases} \Leftrightarrow \begin{cases} u_1 = 0{,}00742\frac{pL^4}{EI} \\ u_2 = -0{,}000519\frac{pL^3}{EI} \end{cases} \quad a)$$

b) Momento fletor da viga em $x = L/2$

$$M(x) = \frac{d^2v}{dx}\cdot EI$$

$$v(x) = N_1\cdot u_1 + N_2\cdot u_2 + N_3\cdot u_3 + N_4\cdot u_4$$

$$v(x) \underset{①}{=} N_3\cdot u_1 + N_4\cdot u_2 = \left(\frac{3x^2}{(L/2)^2} - \frac{2x^3}{(L/2)^3}\right)\times\left(0{,}00742\,\frac{pL^4}{EI}\right) + \left(-\frac{x^2}{(L/2)} + \frac{x^3}{(L/2)^2}\right)\times\left(-0{,}000519\,\frac{pL^3}{EI}\right)$$

$$\frac{dv}{dx} = \left(\frac{6x}{(L/2)^2} - \frac{6x^2}{(L/2)^3}\right)\times 0{,}00742\,\frac{pL^4}{EI} + \left(-\frac{2x}{(L/2)} + \frac{3x^2}{(L/2)^2}\right)\times\left(-0{,}000519\,\frac{pL^3}{EI}\right)$$

$$\frac{d^2v}{dx} = \left(\frac{6}{(L/2)^2} - \frac{12x}{(L/2)^3}\right)\times 0{,}00742\,\frac{pL^4}{EI} + \left(\frac{-2}{(L/2)} + \frac{6x}{(L/2)^2}\right)\times\left(-0{,}000519\,\frac{pL^3}{EI}\right)$$

$$M(x) = \left(\frac{6}{(L/2)^2} - \frac{12x}{(L/2)^3}\right)\times 0{,}00742\,\frac{pL^4}{\cancel{EI}} + \left(\frac{-2}{(L/2)} + \frac{6x}{(L/2)^2}\right)\times\left(-0{,}000519\,\frac{pL^3}{\cancel{EI}}\right)\times\cancel{EI}$$

$$M(x = L/2) = \left(\frac{6}{(L/2)^2} - \frac{12(L/2)}{(L/2)^3}\right)\times 0{,}00742\,pL^4 + \left(\frac{-2}{(L/2)} + \frac{6(L/2)}{(L/2)^2}\right)\times\left(-0{,}000519\,pL^3\right)$$

$$= -0{,}182\,pL^2 \;//$$

c) Rotação aproximada em $x = \frac{3L}{4}$

$\theta(x = L/4)$ → 2º elemento

$\theta(x) = \frac{dv}{dx}$

$v(x)_{②} = N_1 . u_1 + N_2 . u_2 = \left(1 - \frac{3x^2}{(L/2)^2} + \frac{2x^3}{(L/2)^3}\right) \times 0{,}00742 \frac{pL^4}{EI} + \left(x - \frac{2x^2}{(L/2)} + \frac{x^3}{(L/2)^2}\right) \times \left(-0{,}000519 \frac{pL^3}{EI}\right)$

$\theta(x) = \left(\frac{-6x}{(L/2)^2} + \frac{6x^2}{(L/2)^3}\right) \times 0{,}00742 \frac{pL^4}{EI} + \left(1 - \frac{4x}{(L/2)} + \frac{3x^2}{(L/2)^2}\right) \times \left(-0{,}000519 \frac{pL^3}{EI}\right)$

$\theta(x = L/4) = \left(\frac{-6(L/4)}{(L/2)^2} + \frac{6(L/4)^2}{(L/2)^3}\right) \times 0{,}00742 \frac{pL^4}{EI} + \left(1 - \frac{4(L/4)}{(L/2)} + \frac{3(L/4)^2}{(L/2)^2}\right) \times \left(-0{,}000519 \frac{pL^3}{EI}\right) =$

$= -0{,}02213 \frac{pL^3}{EI}$ //

d) Tensão normal da viga a $L/4$

$\sigma = - \frac{M \times y}{I_z}$

$M(x = L/4) = \left(\frac{6}{(L/2)^2} - \frac{12(L/4)}{(L/2)^3}\right) \times 0{,}00742\, pL^4 + \left(\frac{-2}{(L/2)} + \frac{6(L/4)}{(L/2)^2}\right) \times (-0{,}000519\, pL^3)$

$= -0{,}001038\, pL^2$

$\sigma = - \frac{(-0{,}001038 pL^2) \times y}{I} = \frac{0{,}001038\, pL^2 \times y}{I}$ //

e)

$M_{11} = M_{33}^{①} + M_{11}^{②} = \frac{\rho \times A \times (L/2)}{420} \times 156 + \frac{\rho A \times (L/2)}{420} \times 156 = 0{,}371\, \rho AL$

$M_{12} = M_{34}^{①} + M_{12}^{②} = \frac{\rho \times A \times (L/2)}{420} \times (-22) \times (L/2) + \frac{\rho \times A \times (L/2)}{420} \times 22 \times (L/2) = 0$

$M_{22} = M_{44}^{①} + M_{22}^{②} = \frac{\rho A . (L/2)}{420} \times 4 . (L/2)^2 + \frac{\rho A . (L/2)}{420} \times 4 \times (L/2)^2 = 0{,}002381\, \rho . A . L^3$

$|[K] - w^2 [m]| = 0 \iff$

$\iff \left| \frac{EI}{L^3} \begin{bmatrix} 192 & 0 \\ 0 & 16L^2 \end{bmatrix} - w^2 \rho AL \begin{bmatrix} 0{,}371 & 0 \\ 0 & 0{,}002381 L^2 \end{bmatrix} \right| = 0 \iff$

$\lambda = \frac{w^2 \rho A . L . L^3}{EI} = \frac{w^2 . \rho . A . L^4}{EI}$

$$\Leftrightarrow \left| \begin{bmatrix} 192 & 0 \\ 0 & 16L^2 \end{bmatrix} - \lambda \begin{bmatrix} 0{,}371 & 0 \\ 0 & 0{,}00238L^2 \end{bmatrix} \right| = 0 \Leftrightarrow \left| \begin{bmatrix} 192 - 0{,}371\lambda & 0 \\ 0 & 16L^2 \cdot 0{,}00238\lambda L^2 \end{bmatrix} \right| = 0$$

$$\Leftrightarrow (192 - 0{,}371\lambda)(16L^2 - 0{,}00238\lambda L^2) = 0 \Leftrightarrow$$

$$\Leftrightarrow 3072L^2 - 0{,}045715\lambda L^2 - 5{,}936\lambda L^2 + 0{,}00088\lambda^2 L^2 = 0 \Leftrightarrow$$

$$\Leftrightarrow 0{,}00088\lambda^2 - 6{,}393\lambda + 3072 = 0 \Leftrightarrow \lambda = \frac{6{,}393 \pm \sqrt{(6{,}393)^2 - 4 \times 3072 \times (0{,}00088)}}{2 \times 0{,}00088} \Leftrightarrow$$

$$\Leftrightarrow \lambda_1 = 517{,}371 \quad \wedge \quad \lambda_2 = 6747{,}4$$

$$\omega_1 = \sqrt{\frac{\lambda EI}{\rho A L^4}} = \sqrt{\frac{517{,}371 EI}{\rho \times A \times L^4}} = \frac{22{,}746}{L^2} \times \sqrt{\frac{EI}{\rho . A}}$$

$$\omega_2 = \sqrt{\frac{6747{,}4 . EI}{\rho \times A \times L^4}} = \frac{82{,}14}{L^2} \times \sqrt{\frac{EI}{\rho \times A}}$$

Associado a $\lambda_1 : \omega_1$

$$\begin{bmatrix} 192 - 0{,}371(517{,}371) & 0 \\ 0 & 16L^2 - 0{,}00238L^2(517{,}371) \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \end{Bmatrix} = 0 \Leftrightarrow \begin{cases} 0{,}055U_1 + 0U_2 = 0 \\ 0U_1 + 1{,}47681U_2L^2 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} U_1 = \text{qualquer} \\ U_2 = \text{qualquer} \end{cases} \Rightarrow \begin{matrix} U_1 = 0 \\ U_2 = 1 \end{matrix}$$



Associado a $\lambda_2 : \omega_2$

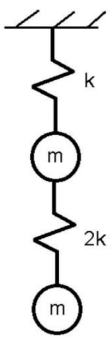
$$\begin{bmatrix} 192 - 0{,}371(6747{,}4) & 0 \\ 0 & 16L^2 - 0{,}00238L^2(6747{,}4) \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \end{Bmatrix} \Leftrightarrow \begin{cases} -2311U_1 + 0U_2 = 0 \\ 0U_1 - 0{,}066U_2L^2 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} U_1 = \text{qualquer} \\ U_2 = \text{qualquer} \end{cases} \Rightarrow \begin{matrix} U_1 = 0 \\ U_2 = 0 \end{matrix}$$



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(14)





Força nas molas com Castigliano

m.c

	1	2
①	-	1
②	1	2

$K_{11} = K_{22}^{①} + K_{11}^{②} = K + 2K = 3K$

$K_{12} = K_{12}^{②} = -2K$

$K_{22} = K_{22}^{②} = 2K$

$$[K] = \begin{bmatrix} 3K & -2K \\ -2K & 2K \end{bmatrix}$$

Princípio Energia potencial mínima

$$U_e = \frac{1}{2} K.u_1^2 + \frac{1}{2}.2K.(u_2-u_1)^2 = \frac{1}{2}\left[K.u_1^2 + 2K(u_1^2 - 2u_1u_2 + u_2^2)\right]$$

$$W = m.g.u_1 + mg.u_2$$

$$\pi = U_e - W = \frac{1}{2}\left[K.u_1^2 + 2K(u_1^2 - 2u_1u_2 + u_2^2)\right] - mg\,u_1 - mg.u_2$$

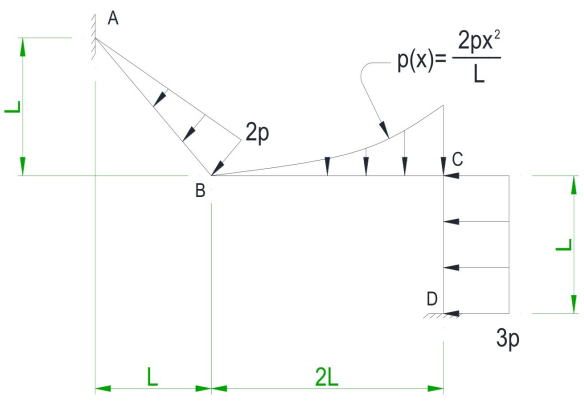
$$\begin{cases} \frac{\partial \pi}{\partial u_1} = 0 \\ \frac{\partial \pi}{\partial u_2} = 0 \end{cases} \Leftrightarrow \begin{cases} Ku_1 + 2Ku_1 - 2Ku_2 = mg \\ 2Ku_2 - 2Ku_1 = mg \end{cases} \Leftrightarrow \begin{cases} 3Ku_1 - 2Ku_2 = mg \\ -2Ku_1 + 2Ku_2 = mg \end{cases}$$

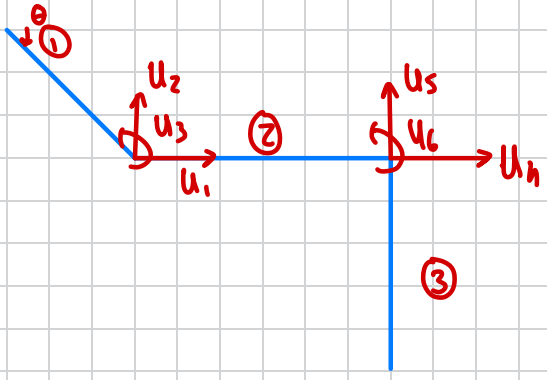
$$\Leftrightarrow \begin{cases} u_1 = \frac{2mg}{K} \\ u_2 = \frac{5mg}{2K} \end{cases}$$

Força na mola ① $\Rightarrow F^{①} = K.u_1 = 2mg$

Força na mola ② $\Rightarrow F^{②} = 2K(u_2-u_1) = 2K\left(\frac{5mg}{2K} - \frac{2mg}{K}\right) \Leftrightarrow F^{②} = mg$

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a) Vetor de forças

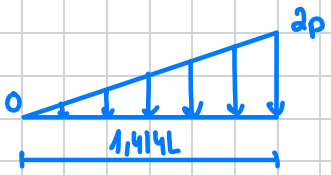
$L^{①} = \sqrt{L^2 + L^2} = 1{,}414L$

$L^{②} = 2L$

$L^{③} = L$

$\theta^{①} = tg^{-1}(L/L) = -45°$

Elemento ①



$f = P_2 \cdot \frac{x}{L} + P_1 \Rightarrow \begin{cases} x = 0 \Rightarrow 0 = P_2 \cdot \frac{0}{1{,}414L} + P_1 \Rightarrow P_1 = 0 \\ x = 1{,}414L \Rightarrow -2p = P_2 \cdot \frac{1{,}414L}{1{,}414L} + 0 \Leftrightarrow P_2 = -2p \end{cases}$

$\{g^{①}\} = \left\{ 0 ;\ \frac{3(-2p) \times (1{,}414L)}{20} ,\ \frac{(-2p)(1{,}414L)^2}{30} ;\ 0 ;\ \frac{7(-2p) \times (1{,}414L)}{20} ;\ -\frac{(-2p)(1{,}414L)^2}{20} \right\}^T$

$$\{g^{①}\} = \begin{Bmatrix} 0 \\ -0{,}4242\,pL \\ -0{,}133\,pL^2 \\ 0 \\ -0{,}99\,pL \\ 0{,}2\,pL^2 \end{Bmatrix}$$

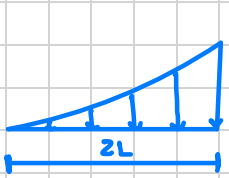
$\{F^{①}\} = [R]^T \times \{g^{①}\} \Leftrightarrow \{F^{①}\} =$

$$\begin{bmatrix} \cos -45 & -\sin -45 & 0 & 0 & 0 & 0 \\ \sin -45 & \cos -45 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos -45 & -\sin -45 & 0 \\ 0 & 0 & 0 & \sin -45 & \cos -45 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ -0{,}4242\,pL \\ -0{,}133\,pL^2 \\ 0 \\ -0{,}99\,pL \\ 0{,}2\,pL^2 \end{Bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \{F^{①}\} = \begin{Bmatrix} -0{,}3\,pL \\ -0{,}3\,pL \\ -0{,}133\,pL^2 \\ -0{,}7\,pL \\ -0{,}7\,pL \\ 0{,}2\,pL^2 \end{Bmatrix}$$

Usam apenas estes

Elemento ②



$\{F^{②}\} = \{0 ;\ F_2^{②} ;\ F_3^{②} ;\ 0 ;\ F_5^{②} ;\ F_6^{②}\}$

$F_2^{②} = -\int_0^{2L} p(x)\, N_1 \cdot dx = -\int_0^{2L} \left(\frac{2px^2}{L}\right)\left(1 - \frac{3x^2}{(2L)^2} + \frac{2x^3}{(2L)^3}\right) \cdot dx = -\int_0^{2L} \frac{2px^2}{L} - \frac{6px^4}{4L^3} + \frac{4px^5}{8L^4} \cdot dx = -1{,}06pL^2$

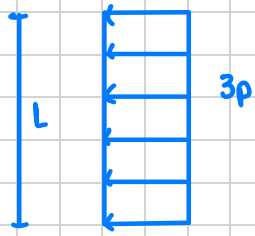
$$F_3^{②} = -\int_0^{2L} p(x)\,N_2\,dx = -\int_0^{2L} \left(\frac{2px^2}{L}\right)\cdot\left(x - \frac{2x^2}{2L} + \frac{x^3}{(2L)^2}\right)\cdot dx = -\int_0^{2L} \frac{2px^3}{L} - \frac{4px^4}{2L^2} + \frac{2px^5}{4L^3}\cdot dx = -0{,}533\,pL^3 \;//$$

$$F_5^{②} = -\int_0^{2L} p(x)N_3\,dx = -\int_0^{2L} \left(\frac{2px^2}{L}\right)\cdot\left(\frac{3x^2}{(2L)^2} - \frac{2x^3}{(2L)^3}\right)dx = -\int_0^{2L} \frac{6px^4}{4L^3} - \frac{4px^5}{8L^4}\cdot dx = -4{,}267\,pL^2 \;//$$

$$F_6^{②} = -\int_0^{2L} p(x)\,N_4\,dx = -\int_0^{2L} \left(\frac{2px^2}{L}\right)\cdot\left(-\frac{x^2}{2L} + \frac{x^3}{(2L)^2}\right)dx = -\int_0^{2L} -\frac{2px^4}{2L^2} + \frac{2px^5}{4L^3}\,dx = 1{,}067\,pL^3 \;//$$

$$\{F^{②}\} = \left\{\begin{array}{c} 0 \\ -1{,}06\,pL^2 \\ -0{,}533\,pL^3 \\ 0 \\ -4{,}267\,pL^2 \\ 1{,}067\,pL^3 \end{array}\right\}$$ Usam todos

Elemento ③



$$\{F^{③}\} = \left\{-\frac{3pL}{2};\ 0;\ -\frac{3pL^2}{12};\ -\frac{3pL}{2};\ 0;\ -\frac{(-3p)L^2}{12}\right\}^T$$

$$\{F^{③}\} = \left\{\begin{array}{c} -1{,}5\,pL \\ 0 \\ -0{,}25\,pL^2 \\ -1{,}25\,pL \\ 0 \\ 0{,}25\,pL^2 \end{array}\right\}$$ Apenas temos estes

Vetor de forças:

$$\{F\} = \left\{\begin{array}{c} -0{,}7pL + 0 \\ -0{,}7pL - 1{,}06pL^2 \\ 0{,}2pL^2 - 0{,}533\,pL^3 \\ 0 - 1{,}5pL \\ -4{,}267\,pL^2 + 0 \\ 1{,}067\,pL^3 - 0{,}25pL^2 \end{array}\right\} \Rightarrow \{F\} = \left\{\begin{array}{c} -0{,}7pL \\ -0{,}7pL + 1{,}06pL^2 \\ 0{,}2\,pL^2 - 0{,}533\,pL^3 \\ -1{,}5pL \\ -4{,}267\,pL^2 \\ 1{,}067\,pL^3 - 0{,}25pL^2 \end{array}\right\} \;//$$